

Human Personality-Based Particle Swarm Optimization: A Trait-Driven Framework for High-Dimensional Heterogeneous Problems

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Abstract

Particle Swarm Optimization (PSO) suffers from premature convergence and loss of diversity, particularly in high-dimensional and multimodal optimization problems. To address these limitations, this paper proposes a novel Human Personality-Based Particle Swarm Optimization (HPPSO) algorithm, which introduces personality-inspired heterogeneous behaviors into swarm particles. In the proposed framework, each particle is assigned a stochastic personality vector composed of four behavioral traits: curiosity, aggressiveness, conformity, and openness. These traits modify particle dynamics to encourage diversified exploration and search behavior. In particular, the openness trait introduces a counterintuitive exploration mechanism in which particles are occasionally attracted toward poorly performing peers, enabling the swarm to escape local optima and explore potentially abandoned regions of the search space.

The proposed algorithm was evaluated on 20 shifted benchmark functions under both 30-dimensional and 1000-dimensional settings, as well as on the CEC2011 benchmark suite and neural network training tasks. Experimental results demonstrated that HPPSO achieves competitive performance against state-of-the-art optimization algorithms, including SHADE, CMAES, and sep-CMAES, while maintaining computational complexity comparable to standard PSO. Convergence analyses further indicates that HPPSO preserves swarm diversity by maintaining multiple promising search trajectories throughout the optimization process.

The results suggest that personality-inspired heterogeneity provides an effective mechanism for enhancing exploration in large-scale optimization problems and offers a new perspective for modeling diverse behaviors in swarm intelligence algorithms.

Keywords: Human personality-based PSO, swarm intelligence, heterogeneous particle swarm optimization, behavior-inspired optimization, high-dimensional optimization, neural network training, shifted benchmark functions, metaheuristic optimization

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1 Introduction and Motivation of the work

Particle Swarm Optimization (PSO) [Kennedy and Eberhart \[1995\]](#) is one of the most widely used metaheuristic algorithms [Jain et al. \[2022\]](#). PSO operates using a swarm of agents, called particles, that collaborate to find an optimal solution to a given problem [Zhang et al. \[2015\]](#). Each particle adjusts its velocity based on two factors: the distance to its own best-known position and the distance to the best-known position in the swarm. In essence, particles learn both from personal experience and social interaction.

Despite the algorithm’s effectiveness, standard PSO assumes that all particles behave identically, with uniform update mechanisms and movement strategies. However, real-world swarms—whether composed of birds, fish, or humans—exhibit diverse behaviors and traits among individuals.

In human societies, for example, individuals vary in personality traits such as confidence, curiosity, sociability, aggressiveness, and risk tolerance. If a swarm were to mimic human behavior, it would need to reflect these differences. Simulating such heterogeneity could yield smarter and more adaptable algorithms, mirroring the continuous evolution and complexity of human decision-making. This motivation led to the development of the Human Personality-Based PSO (HPPSO).

HPPSO is not merely a novel algorithm; it is an extensible framework designed to simulate human personality traits within the swarm. Each particle embodies a distinct set of traits that evolve during the optimization process. The framework is modular and flexible, allowing future researchers to incorporate additional traits as needed. This extensibility is the core reason HPPSO is referred to as a “framework” rather than just an algorithm.

1.1 Advantages of HPPSO

1.1.1 Feasibility for High-Dimensional Problems

Traditional metaheuristic algorithms, including standard PSO, often suffer from the curse of dimensionality [Chen et al. \[2015\]](#), making them inefficient for solving high-dimensional problems due to premature convergence [Jia et al. \[2011\]](#). HPPSO mitigates this issue by introducing fuzzy behavioral settings for each individual particle. This diversity helps the swarm escape local minima and enhances global exploration.

1.1.2 Feasibility for Heterogeneous Problems

Heterogeneous problems—where the optimal solution requires different or diverse values across dimensions—are common in real-world scenarios. In contrast, homogeneous problems are less prevalent, yet traditional metaheuristics often perform better on these simpler tasks. Many standard algorithms tend to converge toward biased or trivial solutions, such as the zero vector, as noted in [Morales-Castañeda et al. \[2025\]](#).

HPPSO addresses this limitation through its inherent diversity. By assigning unique and fuzzy behavioral traits to each particle, the algorithm achieves better adaptability and robustness when solving heterogeneous problems.

2 Literature Review

2.1 Metaheuristic Algorithms

Metaheuristic algorithms are widely recognized as effective approaches for solving complex optimization problems, particularly when traditional methods struggle with nonlinear, multimodal, or non-deterministic problems, or those that are difficult to model mathematically [Turgut et al. \[2023\]](#), [Bandaru and Deb \[2016\]](#).

These algorithms are often inspired by nature and employ probabilistic mechanisms to balance exploration and exploitation, thereby enhancing their ability to find global or near-global optimal solutions [Bandaru and Deb \[2016\]](#).

Among them, Particle Swarm Optimization (PSO), developed by Kennedy and Eberhart [Kennedy and Eberhart \[1995\]](#), stands out due to its simplicity and effectiveness.

2.2 Classical Particle Swarm Optimization (PSO)

Particle Swarm Optimization (PSO), introduced by Kennedy and Eberhart in 1995 [Kennedy and Eberhart \[1995\]](#), simulates the social behavior observed in bird flocking and fish schooling. Each particle in the swarm represents a potential solution and updates its position based on personal experience and information shared by other particles. A particle's movement is governed by its velocity, which is influenced by both its own best-known position and the global best position found by the swarm. Despite its popularity, PSO suffers from limitations such as premature convergence, reduced diversity, and lack of individualized particle behavior [Aote et al. \[2013\]](#).

2.3 Enhancements to PSO

2.3.1 Structural Enhancements

Several researchers have proposed structural modifications to improve PSO performance. These include adopting alternative communication topologies such as ring, star, or fully connected networks [\[Gad, 2022, p. 2534\]](#), and partitioning the swarm into multiple sub-swarms, each exploring different regions of the objective function landscape [\[Gad, 2022, p. 2538\]](#). Yang et al. [Yang et al. \[2023\]](#) developed Random Contrastive Interaction PSO (RCI-PSO), which rethinks particle interaction in high-dimensional search by replacing fixed guidance with random, contrastive, and dynamic peer-based influence. This strategy enables the development of an adaptive communication topology that significantly outperforms other PSO variants.

2.3.2 Behavioral Enhancements

Behavioral enhancements modify how particles update their velocities and positions. For instance, Chaotic PSO introduces controlled randomness to enhance global exploration, while Quantum-behaved PSO integrates principles from quantum mechanics to improve convergence [\[Gad, 2022, p. 2534\]](#). Additionally, adaptive mechanisms for dynamically adjusting parameters have been proposed to better balance exploration and exploitation [Wang et al. \[2021\]](#).

2.3.3 Bio-Inspired Enhancements

Several PSO variants are inspired by specific biological swarms, including Cat Swarm Optimization [Ihsan et al. \[2021\]](#), Bird Mating Optimizer [Arram et al. \[2021\]](#), and the Firefly Algorithm [\[Yang, 2010, p. 96\]](#). These variants aim to incorporate diverse swarm behaviors, thereby expanding the algorithm’s capabilities and promoting further innovation.

2.3.4 Hybrid PSO Algorithms

To mitigate PSO’s limitations such as poor local search and premature convergence, researchers have developed hybrid algorithms that combine PSO with other optimization methods.

For instance, Shi et al. [Shi et al. \[2022\]](#) proposed a PSO-GA hybrid that incorporates genetic operators from Genetic Algorithms (GAs) [Forrest \[1996\]](#), such as crossover and mutation, into the PSO framework.

Similarly, NDWPSO [Qiao et al. \[2024\]](#) integrates elements from Differential Evolution (DE) [Das and Suganthan \[2010\]](#) and the Whale Optimization Algorithm (WOA) [Mirjalili and Lewis \[2016\]](#) to enhance local search capabilities.

Other studies, such as Shaik et al. [Shaik et al. \[2024\]](#), have embedded local search methods like Simulated Annealing (SA) [Pardalos and Mavridou \[2024\]](#) into PSO to prevent premature convergence and facilitate escape from local minima by accepting worse solutions under a temperature-controlled probability.

Recent work by Wang et al. [Wang et al. \[2024\]](#) introduced a multi-swarm PSO that incorporates reinforcement learning (RL) into the search process, opening a new research direction through the integration of learning-based strategies.

Despite these advances, no algorithm to date has incorporated psychological realism, leaving room for the development of hybrid approaches that model human-like variability in behavior.

2.4 Human-Inspired Metaheuristics

Recent trends in metaheuristics have explored human-inspired strategies [\[Leiva et al., 2024, p. 1\]](#). Notable examples include Human behavior-based particle swarm optimization (HPSO) [Liu et al. \[2014\]](#) that adapts learning from the worst particle in the swarm to avoid its location or to move towards it to enhance exploration and postpone the premature convergence, and Human Mental Search (HMS) [Mousavirad and Ebrahimpour-Komleh \[2017\]](#), which simulates human decision-making in bid-space exploration during online auctions; Teaching-Learning-Based Optimization (TLBO) [Rao et al. \[2011\]](#), which models educational interactions between teachers and learners; and Human Learning Optimization (HLO) [Wang et al. \[2014\]](#), which emulates human learning processes. These algorithms emphasize cognitive and social behaviors rather than purely biological mechanisms.

However, most human-inspired algorithms assume behavioral homogeneity, treating all agents as if they behave identically. This assumption contradicts real-world human populations, where individuals exhibit diverse and often unpredictable behavior patterns, thereby limiting the realism and adaptability of such models.

2.5 Personality Traits in Optimization and the Research Gap

In real-world societies, individuals possess diverse personality traits that influence their behavior and decision-making, as described by Goldberg [Goldberg \[2013\]](#). Traits such as openness, risk-taking, curiosity, and sociability shape how individuals act. However, most population-based optimization algorithms fail to model this behavioral diversity effectively. When such traits are incorporated, they are often represented in a rigid, binary manner—for example, an agent is either fully curious or not curious at all.

For instance, in HLO [Wang et al. \[2014\]](#) and TLBO [Rao et al. \[2011\]](#), agents are initially homogeneous. Although role switching is allowed during optimization, it typically involves complete transitions from one role to another (e.g., learner to teacher) rather than exhibiting blended or probabilistic roles (e.g., partially teacher, partially learner). This lack of granularity and stochastic behavior reduces the diversity and realism of the optimization process.

3 Methodology

In this section, we describe the theoretical foundations of the proposed algorithm. Specifically, we define the personality traits and explain their role in enhancing algorithmic performance.

Humans typically do not exhibit absolute personality traits; for example, individuals are rarely entirely introverted or extroverted, aggressive or peaceful. Instead, personality traits are expressed to varying degrees. To simulate this behavior, we model traits as continuous values in the range $[0, 1]$. These values are sampled from a uniform distribution to ensure an unbiased distribution across all traits.

3.1 Personality Representation

Each particle i is associated with a personality vector:

$$\mathbf{z}_i = [\text{curio}_i, \text{conf}_i, \text{agg}_i, \text{open}_i] \in [0, 1]^4 \quad (1)$$

where:

- curio_i represents the tendency of particle i to perform local exploration,
- conf_i represents the degree to which particle i trusts the global best position, thereby controlling the extent of its movement toward the global best,
- agg_i represents the personalized inertia weight assigned to particle i , replacing the constant inertia weight w used in the standard PSO algorithm,
- open_i determines the probability that particle i utilizes information shared by other particles in the swarm.

These traits are initialized independently for each particle in the swarm and sampled from the uniform distribution $\mathcal{U}(0, 1)$. The generation of these traits is governed by Equation 2, which defines the personality matrix representing the diversity of personalities within the swarm.

$$P = [p_{ij}] \in \mathbb{R}^{n_{\text{pop}} \times T}, \quad p_{ij} \sim \mathcal{U}(0, 1) \quad (2)$$

where:

- P is the personality matrix,
- n_{pop} denotes the number of particles in the swarm,
- T denotes the number of personality traits,
- p_{ij} represents the value of the j -th personality trait for the i -th particle, sampled from a uniform distribution over the interval $[0, 1]$.

3.2 Mathematical Modeling of Personality Traits

In this section, we present the mathematical formulation of the personality traits embedded in HPPSO.

3.2.1 Curiosity

Each particle explores its local neighborhood through a Gaussian mutation process with a probability determined by its curiosity trait. The mutation probability is defined as follows:

$$P_{mut}^{(i,d)} = \alpha_{mut} \cdot \text{curio}_i, \quad (3)$$

where α_{mut} denotes the maximum allowable mutation probability for a particle, with values in the interval $[0, 1]$. This parameter controls the overall exploration capability of the swarm and is set to 0.5 by default.

With probability $P_{mut}^{(i,d)}$, particle i updates its position in dimension d according to Equation 4:

$$x_{i,d} \leftarrow x_{i,d} + \mathcal{N}(0, \eta \cdot (ub_d - lb_d)), \quad (4)$$

where $x_{i,d}$ denotes the position of particle i in dimension d , $\mathcal{N}$ represents a Gaussian distribution, and η is a hyperparameter that controls the mutation scale relative to the search-space interval. By default, η is assigned a small value, such as 0.01, to encourage localized random exploration around the current position.

Although curiosity may appear conceptually similar to mutation strategies previously introduced in the PSO literature, such as the approach proposed in Higashi and Iba [2003], the proposed mechanism differs from conventional global mutation approaches. In standard PSO variants incorporating mutation, all particles are typically assigned the same mutation probability, resulting in homogeneous exploration behavior throughout the swarm. In contrast, HPPSO assigns an individualized mutation probability to each particle according to its curiosity trait, producing particles with varying exploration tendencies ranging from exploratory to exploitative behaviors.

Consequently, the swarm becomes heterogeneous, enabling particles to adopt distinct functional roles during the optimization process. This heterogeneity enhances behavioral diversity and reduces the likelihood of premature convergence, a well-known limitation of the standard PSO algorithm. Previous studies on heterogeneous PSO variants have similarly demonstrated the effectiveness of incorporating behavioral diversity into swarm optimization frameworks Yin et al. [2015].

3.2.2 Conformity

Although standard PSO incorporates both personal and global learning components, the personal best positions are not entirely independent of the global best. As iterations progress, particles influenced by the global best tend to update their personal best positions toward similar regions of the search space, i.e., $\mathbf{p}_i \rightarrow \mathbf{g}$. Consequently, the cognitive and social components become increasingly aligned:

$$(\mathbf{p}_i - \mathbf{x}_i^t) \approx (\mathbf{g} - \mathbf{x}_i^t),$$

which increases the correlation between them and reduces swarm diversity, thereby increasing the risk of premature convergence.

To address this limitation, a conformity trait is introduced to regulate the degree of attraction toward the global best relative to the personal best. The velocity update is modified as follows:

$$\mathbf{v}_i^{t+1} = (1 - \text{conf}_i) c_1 \mathbf{r}_1 \odot (\mathbf{p}_i - \mathbf{x}_i^t) + \text{conf}_i c_2 \mathbf{r}_2 \odot (\mathbf{g} - \mathbf{x}_i^t), \quad (5)$$

where $\text{conf}_i \in [0, 1]$ denotes the conformity level of particle i . In this formulation, conformity represents the degree to which a particle is attracted to the global best rather than relying on its personal best.

Particles with high conformity ($\text{conf}_i \rightarrow 1$) place greater emphasis on the global best position, promoting strong social convergence. Conversely, particles with low conformity rely more on their personal best positions, encouraging independent and exploratory behavior.

Even when $\mathbf{p}_i \approx \mathbf{g}$, the effective update can be approximated as:

$$\mathbf{v}_i^{t+1} \approx \boldsymbol{\alpha}_i \odot (\mathbf{g} - \mathbf{x}_i^t),$$

where

$$\boldsymbol{\alpha}_i = (1 - \text{conf}_i) c_1 \mathbf{r}_1 + \text{conf}_i c_2 \mathbf{r}_2.$$

Since conf_i varies across particles, the scaling factor $\boldsymbol{\alpha}_i$ remains heterogeneous, preventing complete alignment of particle velocities. This heterogeneity preserves variance in the swarm dynamics, i.e., $\text{Var}(\mathbf{v}_i^{t+1}) > 0$, and delays the collapse of the swarm distribution. As a result, diverse search trajectories are maintained for longer periods, allowing the swarm to retain exploration capabilities while still benefiting from social learning.

By introducing particle-specific conformity levels, the proposed approach replaces the uniform balance of cognitive and social learning in standard PSO with a heterogeneous attraction mechanism that varies the relative influence of the personal best and global best components across particles. This improves robustness and reduces the likelihood of premature convergence in complex search landscapes.

3.2.3 Aggressiveness

In standard PSO, the inertia weight w controls the contribution of the previous velocity to the current update:

$$\mathbf{v}_i^{t+1} = w \mathbf{v}_i^t + \mathbf{u}_i^t, \quad (6)$$

where $\mathbf{u}_i^t$ denotes the combined cognitive and social components. The inertia weight regulates the trade-off between exploration and exploitation: larger values promote global exploration, while smaller values favor local refinement.

In HPPSO, the inertia weight is replaced by a particle-specific aggressiveness trait:

$$\mathbf{v}_i^{t+1} = (0.4 + 0.5 \text{agg}_i) \mathbf{v}_i^t + \mathbf{u}_i^t, \quad (7)$$

where $\text{agg}_i \in [0, 1]$ defines the aggressiveness level of particle i . This induces a bounded inertia weight:

$$w_i \in [0.4, 0.9],$$

ensuring controlled variation in momentum.

This formulation introduces heterogeneous momentum dynamics across particles. High-aggressiveness particles ($\text{agg}_i \rightarrow 1$) exhibit stronger velocity persistence:

$$\mathbf{v}_i^{t+1} \approx 0.9 \mathbf{v}_i^t + \mathbf{u}_i^t,$$

supporting long-range exploration and traversal of flat or deceptive regions. In contrast, low-aggressiveness particles behave with stronger damping:

$$\mathbf{v}_i^{t+1} \approx 0.4 \mathbf{v}_i^t + \mathbf{u}_i^t,$$

favoring local refinement.

From a dynamical perspective, the update can be modeled as a first-order autoregressive process:

$$\mathbf{v}_i^{t+1} = w_i \mathbf{v}_i^t + \mathbf{u}_i^t,$$

with steady-state variance:

$$\text{Var}(\mathbf{v}_i) \propto \frac{\text{Var}(\mathbf{u}_i)}{1 - w_i^2}.$$

Since w_i varies across particles, velocity variance differs within the swarm. Particles with larger w_i explore wider regions, while those with smaller w_i concentrate on local refinement.

This heterogeneity prevents uniform swarm contraction and preserves a range of motion scales, enabling simultaneous exploration and exploitation without relying on a globally scheduled inertia parameter.

By converting inertia into a particle-specific trait, HPPSO introduces diversity in trajectory persistence, improving search robustness and reducing the likelihood of premature convergence in complex multimodal landscapes.

3.2.4 Openness

In standard PSO, particles primarily exploit information from the global best, with no direct interaction among peers. This limits the ability of the swarm to leverage distributed knowledge across individuals. Several PSO variants address this limitation through alternative communication schemes, such as comprehensive learning PSO (CLPSO) [Liang et al. \[2006\]](#), which enables learning from all the peers history in the swarm. In HPPSO, the degree of inter-particle information sharing is made particle-specific, leading to a heterogeneous social interaction structure.

The proposed Openness operator introduces explicit peer-to-peer interaction. A particle selects a random peer from the swarm and interacts with it only if the peer has a worse current fitness value. In addition to the global and personal best influences, this mechanism allows particles to incorporate information from non-optimal peers, potentially reactivating previously explored regions of the search space. Although such peers may currently exhibit inferior performance, their historical

trajectories may still provide useful exploratory signals that are otherwise absent in standard PSO, where updates are typically biased toward superior solutions.

The mathematical formulation of this operator for particle i is given by:

$$\mathbf{v}_i^{t+1} = \mathbf{v}_i^{t+1} + \lambda(\mathbf{p}_j - \mathbf{x}_i^t), \quad (8)$$

where $\lambda > 0$ controls the interaction strength, $\mathbf{x}_i^t$ denotes the current position of particle i , and $\mathbf{p}_j$ is the personal best position of a randomly selected particle $j \neq i$ satisfying:

$$f(\mathbf{x}_j) > f(\mathbf{x}_i),$$

indicating that particle j performs worse than particle i at iteration t .

This mechanism enables the swarm to revisit regions of the search space that are typically under-explored or abandoned in conventional PSO variants. From an information-theoretic perspective, the operator introduces an additional source of variability into the velocity update.

$$\mathbf{v}_i^{t+1} \leftarrow (\mathbf{p}_i - \mathbf{x}_i^t) \cup (\mathbf{g} - \mathbf{x}_i^t) \cup (\mathbf{p}_j - \mathbf{x}_i^t).$$

Consequently, the conditional uncertainty of the velocity update increases:

$$H(\mathbf{v}_i^{t+1} \mid \mathbf{x}_i^t) \uparrow,$$

which reflects greater stochastic diversity in particle trajectories. This increased variability can be beneficial in multimodal optimization problems, as it improves the ability of the swarm to escape local optima.

3.3 HPPSO General Algorithm

This subsection presents the complete formulation of the HPPSO algorithm. Figure 1 illustrates the proposed HPPSO framework, where Z_i represents the personality vector of particle i , consisting of curiosity, conformity, aggressiveness, and openness traits that modulate particle update dynamics.

Some personality configurations may become ineffective for a given optimization landscape. To improve adaptability, a stagnation monitoring mechanism is introduced to identify particles whose personal best solutions do not improve over T_s iterations, where $T_s = 10$ by default. Once stagnation is detected, the corresponding personality vector is randomly reinitialized to encourage exploration of alternative search behaviors.

The complete HPPSO algorithm is presented in Algorithm 1. The algorithm takes the following inputs and produces the following outputs.

Inputs:

- Objective function $f(\cdot)$
- Problem dimensionality d
- Population size n
- Lower and upper search bounds **lb** and **ub**
- Maximum number of iterations T
- Hyperparameters: c_1 , c_2 , η , α_{mut} , θ , λ , and T_s

Outputs:

- Best solution $\mathbf{g}$ found
- Best cost $f(\mathbf{g})$
- History of best costs per iteration (optional)

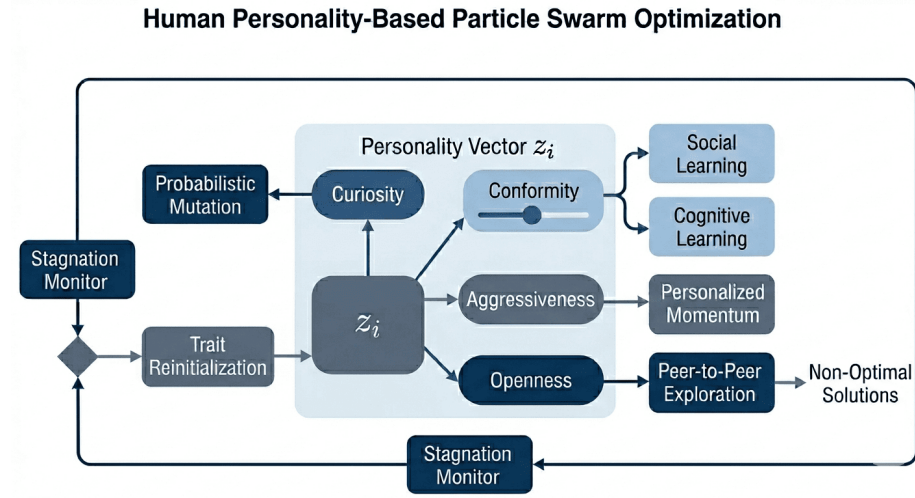


Figure 1: Overview of the proposed HPPSO framework. Z_i represents the personality vector of particle i , composed of curiosity, conformity, aggressiveness, and openness traits that modulate particle update dynamics. A stagnation monitoring mechanism reinitializes the personality vector of stagnant particles to preserve swarm diversity.

The algorithm begins by initializing a random swarm of particles. Each particle is assigned a random personality vector, and its velocity is initialized to zero. A maximum number of iterations is specified.

At each iteration, the fitness of all particles is evaluated, and the global best position (gbest) is identified. Each particle is then updated according to its associated personality-driven mechanism, introducing heterogeneous stochastic behavior across the swarm.

Particles with high openness incorporate an additional attraction component in their velocity update, guiding them slightly toward randomly selected lower-performing particles based on their personal best (pbest) values. To prevent velocity explosion, velocity clamping is applied, which is a standard stabilization technique in PSO variants. Particles with higher curiosity levels are more likely to undergo local perturbations around their current positions.

After position updates, particles are re-evaluated, and both personal best (pbest) and global best (gbest) values are updated accordingly. The personality vectors of stagnating particles, defined as those that fail to improve their pbest over a predefined number of iterations, are reinitialized to encourage exploration of alternative behavioral strategies.

Upon termination of the iteration process, the final global best solution is returned as the optimal solution found by the algorithm.

Algorithm 1 Human Personality-Based Particle Swarm Optimization (HPPSO)

```
1: Initialize  $\mathbf{x}_i \sim \mathcal{U}(\mathbf{lb}, \mathbf{ub})$ ,  $\mathbf{v}_i \leftarrow \mathbf{0}$ 
2: Evaluate  $f(\mathbf{x}_i)$ 
3: Set  $\mathbf{p}_i \leftarrow \mathbf{x}_i$  and initialize  $\mathbf{g}$ 
4: Initialize personality traits and stagnation counters  $s_i \leftarrow 0$ 
5: for  $t = 1$  to  $T$  do
6:   for each particle  $i$  do
7:     Sample  $\mathbf{r}_{1,i}, \mathbf{r}_{2,i} \sim \mathcal{U}(0, 1)^d$ 
8:      $\mathbf{v}_i \leftarrow (0.4 + 0.5\text{agg}_i)\mathbf{v}_i + (1 - \text{conf}_i)c_1\mathbf{r}_{1,i} \odot (\mathbf{p}_i - \mathbf{x}_i) + \text{conf}_i c_2\mathbf{r}_{2,i} \odot (\mathbf{g} - \mathbf{x}_i)$ 
9:     if  $\text{open}_i > \theta$  then
10:       Select  $j \neq i$  with  $f(\mathbf{x}_j) > f(\mathbf{x}_i)$ 
11:       if such  $j$  exists then
12:          $\mathbf{v}_i \leftarrow \mathbf{v}_i + \lambda(\mathbf{p}_j - \mathbf{x}_i)$ 
13:       end if
14:     end if
15:   end for
16:    $\mathbf{v}_i \leftarrow \text{clip}(\mathbf{v}_i, -\mathbf{v}_{\max}, \mathbf{v}_{\max})$ 
17:    $\mathbf{x}_i \leftarrow \mathbf{x}_i + \mathbf{v}_i$ 
18:   for each particle  $i$  and dimension  $k$  do
19:     Sample  $u \sim \mathcal{U}(0, 1)$ 
20:     if  $u < \alpha_{\text{mut}} \times \text{curio}_i$  then
21:        $\mathbf{x}_{i,k} \leftarrow \mathbf{x}_{i,k} + \mathcal{N}(0, \eta(\mathbf{ub}_k - \mathbf{lb}_k))$ 
22:     end if
23:   end for
24:   Project  $\mathbf{x}_i$  onto  $[\mathbf{lb}, \mathbf{ub}]$ 
25:   Evaluate  $f(\mathbf{x}_i)$ 
26:   for each particle  $i$  do
27:     if  $f(\mathbf{x}_i) < f(\mathbf{p}_i)$  then
28:        $\mathbf{p}_i \leftarrow \mathbf{x}_i$ 
29:     end if
30:   end for
31:   Update  $\mathbf{g}$ 
32:   for each particle  $i$  do
33:     if  $f(\mathbf{p}_i^{(t)}) = f(\mathbf{p}_i^{(t-1)})$  then
34:        $s_i \leftarrow s_i + 1$ 
35:     else
36:        $s_i \leftarrow 0$ 
37:     end if
38:     if  $s_i \geq T_s$  then
39:       Reinitialize personality traits
40:        $s_i \leftarrow 0$ 
41:     end if
42:   end for
43: end for
44: return  $\mathbf{g}, f(\mathbf{g})$ 
```

4 Computational Complexity of HPPSO

4.1 Time Complexity

Let n denote the population size, d the dimensionality of the search space, and T the number of iterations.

The computational cost of HPPSO is dominated by particle-state updates, fitness evaluation, mutation operations, and social interaction mechanisms.

Initialization The initialization of particle positions, velocities, personality traits, and auxiliary variables requires:

$$\mathcal{O}(nd)$$

Particle Updates Velocity and position updates consist of element-wise vector operations over all particles and dimensions, yielding:

$$\mathcal{O}(nd)$$

Fitness Evaluation Let C_f denote the computational cost of evaluating the objective function for a single particle. The total evaluation cost is:

$$\mathcal{O}(nC_f)$$

For optimization problems where $C_f = \mathcal{O}(d)$, the resulting complexity becomes:

$$\mathcal{O}(nd)$$

Curiosity-Driven Mutation The mutation mechanism perturbs particle coordinates across all dimensions, resulting in:

$$\mathcal{O}(nd)$$

Social Interaction Mechanism The social interaction mechanism operates over the particle population and contributes:

$$\mathcal{O}(n)$$

Stagnation Handling Updating stagnation counters and personality adaptation variables requires:

$$\mathcal{O}(n)$$

Total Time Complexity The dominant computational terms scale linearly with both population size and dimensionality. Therefore, the overall time complexity per iteration is:

$$\mathcal{O}(nd) \tag{9}$$

Over T iterations, the total computational complexity is:

$$\mathcal{O}(Tnd) \tag{10}$$

4.2 Space Complexity

The space complexity of HPPSO is determined by the storage of particle states and auxiliary variables.

- Particle positions: $\mathcal{O}(nd)$
- Particle velocities: $\mathcal{O}(nd)$
- Personal best positions: $\mathcal{O}(nd)$
- Fitness values: $\mathcal{O}(n)$
- Global best position: $\mathcal{O}(d)$
- Personality traits: $\mathcal{O}(n)$
- Stagnation counters: $\mathcal{O}(n)$

Accordingly, the total space complexity is:

$$\mathcal{O}(nd) \tag{11}$$

Thus, HPPSO preserves the asymptotic memory complexity of conventional particle swarm optimization while incorporating additional personality-driven adaptation mechanisms.

5 Convergence Analysis of HPPSO

HPPSO defines a stochastic population process with persistent random perturbations arising from personality- and interaction-based mechanisms. Unlike classical PSO variants that may employ diminishing stochasticity, HPPSO maintains non-vanishing randomness throughout the search process.

Let $\{\mathbf{X}(t)\}_{t \geq 0}$ denote the swarm state at iteration t , where $\mathbf{X}(t) = \{\mathbf{x}_i(t)\}_{i=1}^n$. The evolution of the system forms a time-homogeneous Markov process, since $\mathbf{X}(t+1)$ depends only on $\mathbf{X}(t)$ and independent random variables.

5.1 Boundedness of the State Space

Velocity vectors are constrained as

$$\mathbf{v}_i(t) \in [-\mathbf{v}_{\max}, \mathbf{v}_{\max}],$$

and positions evolve within a compact domain

$$\mathbf{x}_i(t) \in [\mathbf{lb}, \mathbf{ub}].$$

Hence, the state space is compact, and all trajectories of $\{\mathbf{X}(t)\}$ remain bounded.

5.2 Persistent Stochastic Perturbations

HPPSO incorporates multiple sources of stochasticity, including:

- mutation driven by curiosity,
- stochastic interaction and learning coefficients,
- random peer selection in the social interaction mechanism,
- reinitialization of personality traits under stagnation.

These mechanisms induce a strictly positive conditional variance in the state transitions:

$$\text{Var}[\mathbf{x}_i(t+1) \mid \mathbf{x}_i(t)] > 0.$$

5.3 Best-So-Far Convergence

While individual particle trajectories do not necessarily converge, the best-so-far sequences satisfy monotonic improvement:

$$\mathbb{E}[f(\mathbf{p}_i(t+1))] \leq \mathbb{E}[f(\mathbf{p}_i(t))], \quad \mathbb{E}[f(\mathbf{g}(t+1))] \leq \mathbb{E}[f(\mathbf{g}(t))].$$

Since these sequences are bounded below by the optimal objective value $f(\mathbf{x}^*)$, they converge in expectation to finite limits:

$$\lim_{t \rightarrow \infty} \mathbb{E}[f(\mathbf{g}(t))] = f_\infty.$$

5.4 Asymptotic Exploration Behavior

Due to persistent stochasticity, the swarm does not collapse to a single point. Instead, it maintains sustained exploration of the feasible domain. For any neighborhood of the global optimum region,

$$\mathcal{S}_\epsilon = \{\mathbf{x} \mid f(\mathbf{x}) \leq f(\mathbf{x}^*) + \epsilon\},$$

there exists a non-zero probability that particles visit this region in the long run:

$$\limsup_{t \rightarrow \infty} \mathbb{P}(\mathbf{x}_i(t) \in \mathcal{S}_\epsilon) > 0.$$

5.5 Discussion

HPPSO should therefore be interpreted as a stochastic search process characterized by bounded trajectories and convergence of the best-so-far objective sequence, rather than convergence of individual particle states. Persistent stochasticity prevents collapse of the swarm and sustains exploration throughout the optimization process.

6 Experiments and Results

This section evaluates the HPPSO algorithm. First, an ablation and sensitivity analysis is conducted to examine the influence of individual personality traits on algorithmic performance. In the ablation study, selected traits are disabled, while the sensitivity analysis systematically varies trait-related hyperparameters. This combined methodology enables assessment of both the contribution of individual components and the robustness of HPPSO with respect to parameter variations.

Subsequently, HPPSO is benchmarked against several state-of-the-art optimization algorithms on a set of test problems, including low-, medium-, and high-dimensional benchmark functions, the CEC 2011 benchmark suite, and a real-world neural network training task. The resulting quantitative performance metrics are analyzed to evaluate the effectiveness of the proposed method.

6.1 Component Analysis and Hyperparameter Sensitivity

This subsection investigates the contribution of individual personality traits in the proposed HPPSO algorithm and evaluates the sensitivity of key hyperparameters. Standard benchmark functions with diverse landscape characteristics are employed to isolate the effects of individual components and to determine whether each trait contributes meaningfully to search performance.

The considered benchmark functions include Sphere, Rosenbrock, Ackley, and Levy, selected due to their distinct properties, such as convexity, ill-conditioning, multimodality, and ruggedness. For each experiment, individual personality traits are isolated and evaluated independently on the selected problems.

To ensure statistical reliability, all experiments are repeated over five independent random seeds to reduce initialization bias. The reported results are obtained by averaging convergence curves across seeds at each iteration, providing a consistent measure of both convergence behavior and final objective performance.

For evaluation purposes, all benchmark functions are shifted by random vectors to prevent bias toward trivial optima or center-based solutions. The same shifting vectors are applied consistently across all algorithms to ensure a fair comparison.

The experiments are conducted in a 100-dimensional setting to assess performance on medium-to-high-dimensional problems. The population size is set to 30, and the number of iterations is 2000. For PSO-based comparisons, the inertia weight is fixed at 0.65.

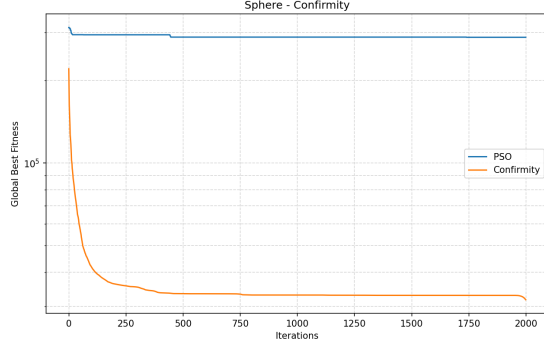
6.1.1 Conformity Component Analysis

In this experiment, standard PSO is compared with a conformity-enhanced PSO variant in which all personality traits except conformity are disabled. The objective is to isolate the effect of the conformity mechanism on optimization performance. All experiments are conducted using five independent random seeds.

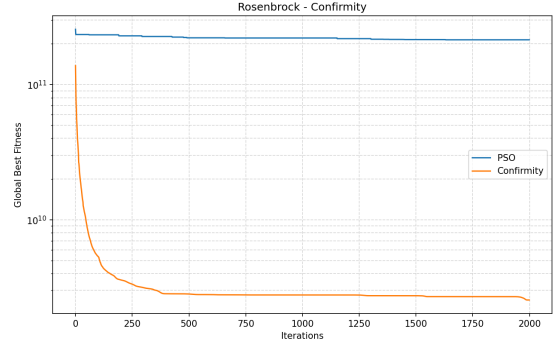
Figure 2 presents the mean convergence curves of standard PSO and the conformity-enhanced variant across four benchmark functions in a 100-dimensional setting.

The results indicate that the conformity trait improves optimization performance across both unimodal and multimodal benchmark functions. This suggests that introducing randomized attraction toward the global best position positively influences the search dynamics of the swarm.

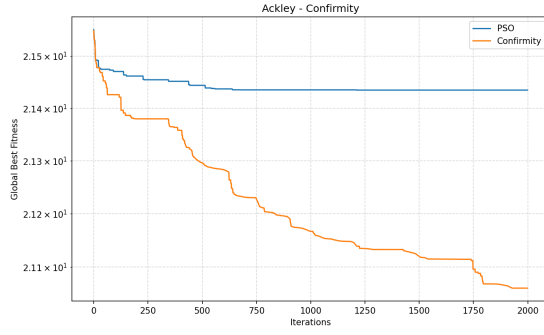
Although the observed improvement is consistent across the considered benchmark problems, further theoretical and experimental investigation is required to establish a rigorous explanation for this behavior.



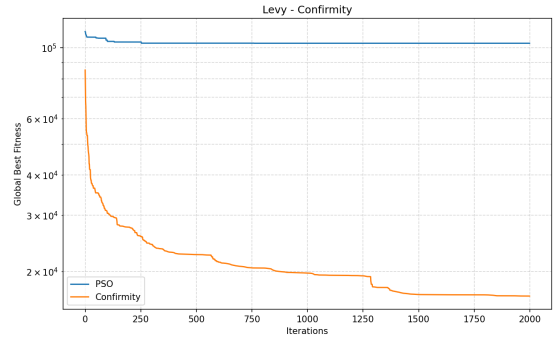
(a) Shifted Sphere



(b) Shifted Rosenbrock



(c) Shifted Ackley



(d) Shifted Levy

Figure 2: Conformity component analysis of HPPSO: mean convergence curves over five random seeds on 100-dimensional shifted benchmark functions.

6.1.2 Aggressiveness Component Analysis

The same experimental setting used in the conformity analysis is adopted in this study. Figure 3 presents the convergence behavior of standard PSO and the aggressiveness-enhanced variant. In addition, comparisons are conducted against PSO-RIW proposed in [Eberhart and Shi \[2001\]](#), a PSO variant employing randomized inertia weights without particle-specific personalization.

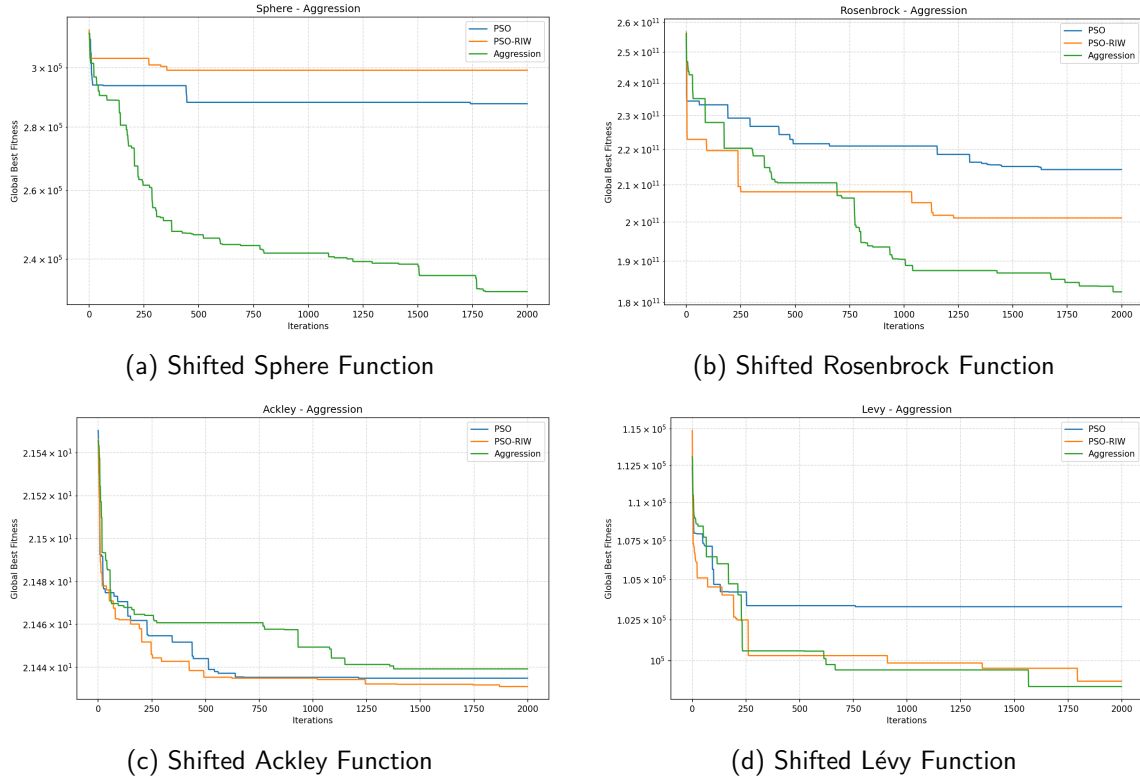


Figure 3: Effect of the aggressiveness trait on convergence behavior across 100-dimensional shifted benchmark functions. Mean convergence curves averaged over five independent runs.

The aggressiveness component improves optimization performance on several benchmark problems, particularly on the shifted Sphere function, where faster convergence and lower final objective values are observed. Improvements are also visible on Rosenbrock and Levy, although the performance gain is less pronounced for highly multimodal problems.

For the Ackley function, all compared variants exhibit similar behavior and tend to become trapped in local minima, indicating that the aggressiveness mechanism alone does not guarantee improved performance on highly deceptive landscapes.

Compared with PSO-RIW, the personalized inertia mechanism exhibits improved empirical performance on several benchmark problems. While randomized inertia weights introduce additional stochasticity into the velocity updates, particle-specific inertia values preserve consistent momentum patterns for individual particles across iterations. This may lead to more distinct search trajectories within the swarm and could explain the observed improvement in optimization performance on both convex and non-convex benchmark functions.

6.1.3 Curiosity Component and Sensitivity Analysis

The same experimental setting used in the conformity analysis is adopted in this study. Figure 4 presents the convergence behavior of PSO with and without the curiosity component. In addition,

tion, comparisons are conducted against PSO-M Higashi and Iba [2003], a variant incorporating non-personalized mutation. The effect of the hyperparameter $\eta \in [0, 1]$ is also analyzed. In this experiment, standard PSO corresponds to the configuration $\eta = 0$.

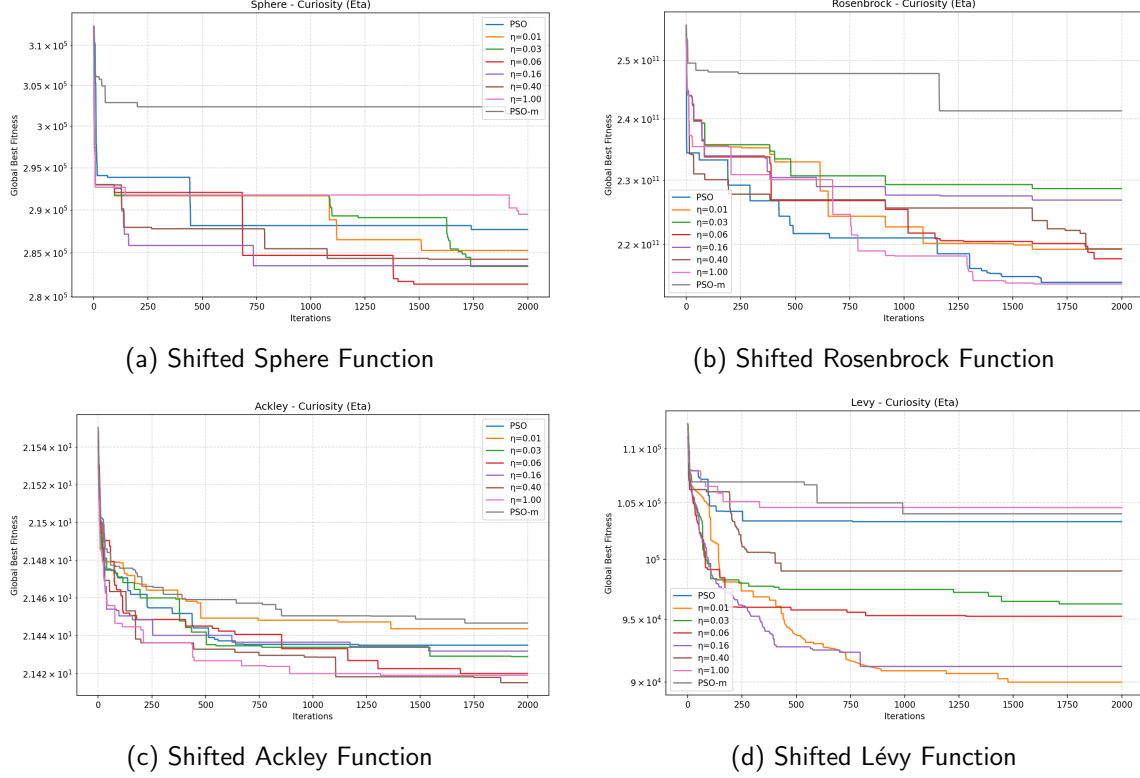


Figure 4: Ablation study of the curiosity trait and hyperparameter sensitivity analysis of η across 100-dimensional shifted benchmark functions. Mean convergence curves are averaged over five independent runs.

The results indicate that PSO and PSO-M exhibit the worst performance among the compared methods, followed by the curiosity-enhanced variant. In addition, performance decreases as η increases, suggesting that lower values of η are more effective for the considered benchmark problems.

The sensitivity analysis further shows that the algorithm consistently improves performance across the tested range $\eta \in [0, 1]$. This indicates that the curiosity component is relatively robust to variations in η , leading to only limited sensitivity with respect to this hyperparameter.

6.1.4 Openness Component and Sensitivity Analysis

The same experimental setting employed in the conformity analysis is adopted in the present experiment. Figure 4 illustrates the convergence behavior of PSO both with and without the proposed social interaction mechanism. The terms *openness* and *socialism* are used interchangeably to denote the same behavioral operator in HPPSO.

In this context, the terms *social interaction* and *openness* are used interchangeably. Accordingly, the parameter *SocThr* corresponds to the threshold parameter θ . The standard PSO configuration corresponds to $SocThr = 1$, where $SocThr \in [0, 1]$.

Additionally, the effect of the social interaction strength hyperparameter $\lambda \in [0, \infty)$ is investigated.

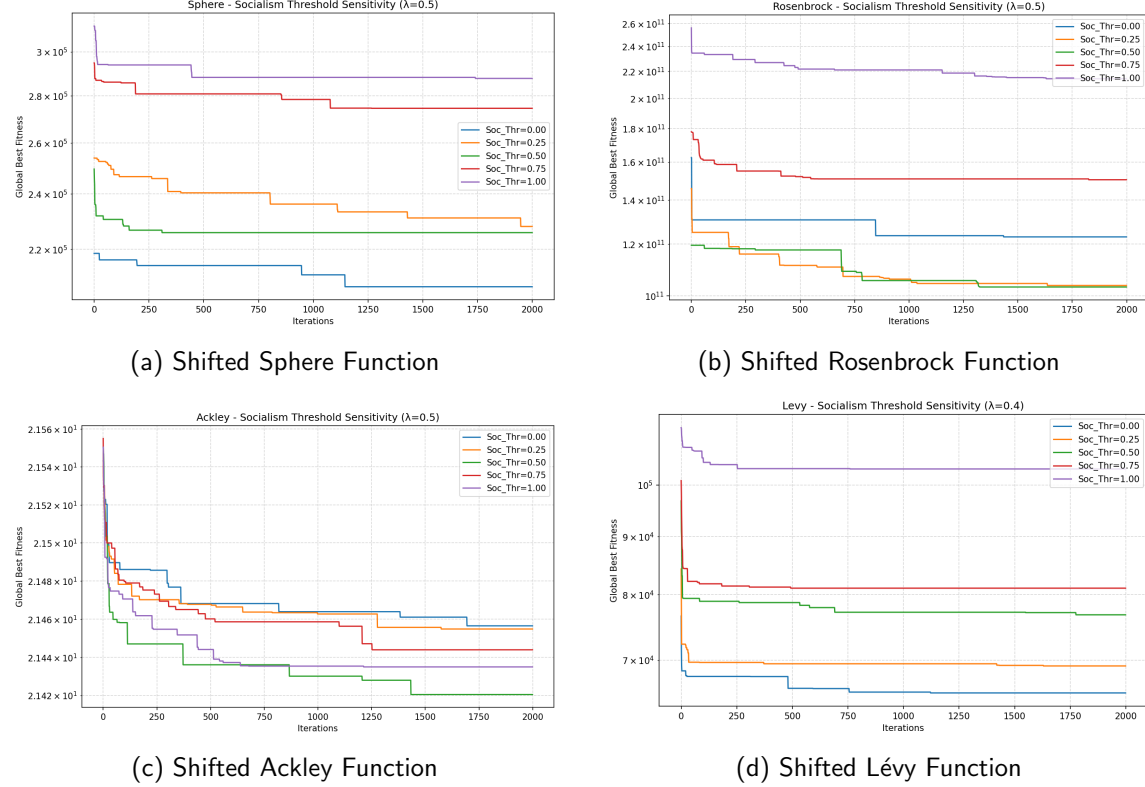


Figure 5: Ablation study of the openness operator and sensitivity analysis of the openness threshold parameter θ (also referred to as *socThr*) on 100-dimensional shifted benchmark functions. Mean convergence curves are averaged over five independent runs.

The results indicate that PSO without the social interaction component exhibits the weakest performance among the compared configurations. All tested threshold values outperform the baseline PSO, indicating consistent improvement across the considered range. In addition, performance is not highly sensitive to variations in the threshold parameter, suggesting that this hyperparameter is relatively easy to tune.

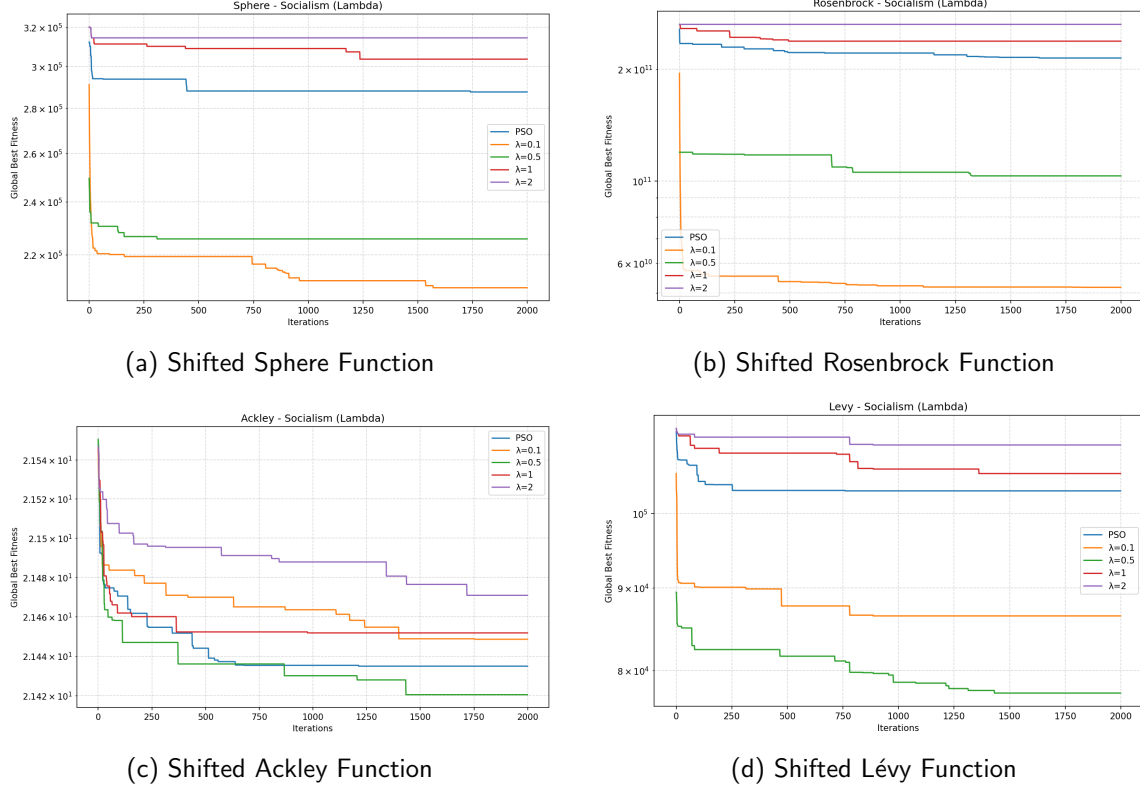


Figure 6: Ablation study of the openness operator and sensitivity analysis of the hyperparameter λ (associated with the *socialism/openness* configuration) on 100-dimensional shifted benchmark functions. Mean convergence curves are averaged over five independent runs.

The hyperparameter λ shows improved performance primarily for values below 1, whereas higher values lead to degraded performance, as shown in Figure 6. This empirical observation suggests that the influence strength of social interaction must be carefully controlled. A more formal analysis of this behavior is left for future work.

Overall, the component analysis indicates that incorporating the social interaction mechanism improves performance across the tested benchmarks, while the sensitivity analysis suggests limited dependence on the threshold parameter.

6.2 Benchmarking

This section presents the experimental evaluation of HPPSO. The algorithm is tested on 20 high-dimensional heterogeneous benchmark functions and compared against several metaheuristic algorithms, namely PSO, PSO-M, PSO-RIW, GWO, CSA, CMAES, SHADE, and GA-MPC. In addition, evaluations are conducted on the CEC2011 benchmark suite. Finally, to demonstrate applicability to real-world problems, experiments are performed on neural network training tasks.

The selected comparison algorithms are described as follows:

1. **Particle Swarm Optimization (PSO) and variants:** PSO [Kennedy and Eberhart \[1995\]](#) is used as the primary baseline for swarm-based optimization. Two variants are included: PSO-M [Higashi and Iba \[2003\]](#), which applies Gaussian mutation uniformly across particles, and PSO-RIW [Eberhart and Shi \[2001\]](#), which incorporates randomly varying inertia weights at each iteration.
2. **CMAES:** The Covariance Matrix Adaptation Evolution Strategy [Ostermeier et al. \[1994\]](#) is a strong baseline for low- and medium-dimensional continuous optimization problems. Its variant, SEP-CMAES [Ros and Hansen \[2008\]](#), is included for high-dimensional settings.
3. **Circle Search Algorithm (CSA):** CSA is a relatively recent metaheuristic proposed in 2022 [Qais et al. \[2022\]](#). It has been shown to perform competitively on several benchmark problems and is included to assess robustness on heterogeneous test functions.
4. **Grey Wolf Optimizer (GWO):** GWO, introduced in 2014 [Mirjalili et al. \[2014\]](#), is a widely used swarm-based optimization method. Despite mixed findings in the literature that question its methodological assumptions [Camacho-Villalón et al. \[2023\]](#), it remains a widely used baseline due to its simplicity and competitive empirical performance.
5. **Success-History Based Parameter Adaptation for Differential Evolution (SHADE)** [Tanabe and Fukunaga \[2013\]](#): SHADE is a differential evolution variant with strong performance on multimodal optimization problems.
6. **Genetic Algorithm with Multi-Parent Crossover (GA-MPC)** [Elsayed et al. \[2011\]](#): GA-MPC is the winner of the CEC2011 competition and serves as a strong evolutionary baseline.

6.2.1 Experiments on Mathematical Functions

HPPSO was evaluated on 20 mathematical benchmark functions commonly used for assessing optimization algorithms. Since the proposed method is designed to address heterogeneous optimization problems, standard benchmark formulations may introduce evaluation bias due to their homogeneous optima, which are typically centered at the origin.

To mitigate this issue, each benchmark function was modified using a randomly generated shift vector applied independently to each dimension. The same shift vector was used across all algorithms for a given benchmark function to ensure fair comparison, while different benchmark functions used different shift vectors. All shift values were uniformly sampled from the search interval $[-100, 100]$.

This modification is motivated by two considerations. First, many benchmark functions possess a global optimum at $\mathbf{x} = \mathbf{0}$, which may favor algorithms exhibiting implicit bias toward origin-centered solutions [Morales-Castañeda et al. \[2025\]](#). Second, homogeneous optima such as $[0, 0, \dots, 0]$ may be trivially discovered by certain optimization strategies, particularly under homogeneous initialization schemes.

Due to the high dimensionality of the experiments, which include problems up to 1000 dimensions, the complete shift vectors are omitted from the main text and provided as supplementary material.

The benchmark functions used in the experiments are listed in Table 1.

Table 1: Shifted Benchmark Functions Used in the Experimental Evaluation

Function	Formula	Domain	Global Min.
Sphere (f_1)	$\sum z_i^2$	$[-100, 100]$	0
Schwefel 2.22 (f_2)	$\sum z_i + \prod z_i $	$[-100, 100]$	0
Zakharov (f_3)	$\sum z_i^2 + (\sum \frac{i}{2} z_i)^2 + (\sum \frac{i}{2} z_i)^4$	$[-100, 100]$	0
Rosenbrock (f_4)	$\sum [100(z_{i+1} - z_i^2)^2 + (z_i - 1)^2]$	$[-100, 100]$	0
Powell Sum (f_5)	$\sum z_i ^{2+\frac{i}{d}}$	$[-100, 100]$	0
Rastrigin (f_6)	$\sum (z_i^2 - 10 \cos(2\pi z_i) + 10)$	$[-100, 100]$	0
Ackley (f_7)	$-20e^{-0.2\sqrt{\frac{1}{d}\sum z_i^2}} - e^{\frac{1}{d}\sum \cos(2\pi z_i)} + 20 + e$	$[-100, 100]$	0
Griewank (f_8)	$\frac{1}{4000} \sum z_i^2 - \prod \cos\left(\frac{z_i}{\sqrt{i}}\right) + 1$	$[-100, 100]$	0
Weierstrass (f_9)	$\sum_{i=1}^d \sum_{k=0}^K a^k \cos(2\pi b^k (z_i + 0.5)) - d \sum_{k=0}^K a^k \cos(2\pi b^k \cdot 0.5)$	$[-100, 100]$	0
Levy (f_{10})	$\sin^2(\pi w_1) + \sum (w_i - 1)^2 [1 + 10 \sin^2(\pi w_i + 1)] + (w_d - 1)^2 [1 + \sin^2(2\pi w_d)]$	$[-100, 100]$	0
Schwefel (f_{11})	$418.9829d - \sum z_i \sin(\sqrt{ z_i })$	$[-100, 100]$	0
Styblinski-Tang (f_{12})	$\frac{1}{2} \sum (z_i^4 - 16z_i^2 + 5z_i)$	$[-100, 100]$	$\approx -39.165d$
Michalewicz (f_{13})	$-\sum \sin(z_i) \sin^{2m}\left(\frac{iz_i^2}{\pi}\right)$	$[-100, 100]$	$\approx -d$
Alpine (f_{14})	$\sum z_i \sin(z_i) + 0.1z_i $	$[-100, 100]$	0
HappyCat (f_{15})	$ \sum z_i^2 - d ^{1/4} + \frac{0.5 \sum z_i^2 + \sum z_i}{d} + 0.5$	$[-100, 100]$	0
HGBat (f_{16})	$ (\sum z_i^2)^2 - (\sum z_i)^2 ^{1/2} + \frac{0.5 \sum z_i^2 + \sum z_i}{d} + 0.5$	$[-100, 100]$	0
Bent Cigar (f_{17})	$z_1^2 + 10^6 \sum_{i=2}^d z_i^2$	$[-100, 100]$	0
Discus (f_{18})	$10^6 z_1^2 + \sum_{i=2}^d z_i^2$	$[-100, 100]$	0
Penalized (f_{19})	$\frac{\pi}{d}[\dots] + \sum u(z_i)$	$[-100, 100]$	0
Zakharov Variant (f_{20})	$(\sum z_i) \exp(\sum \sin(z_i^2))$	$[-100, 100]$	0

All benchmark functions are evaluated in shifted form, where

$$\mathbf{z} = \mathbf{x} - \mathbf{o},$$

and $\mathbf{o}$ denotes the shift vector defined in the previous subsection.

To improve numerical robustness in high-dimensional settings, stabilization techniques such as clipping and log-domain computations are applied where necessary. Implementation details and shift vectors are provided in the supplementary material to ensure reproducibility of the reported results.

Experimental Setup on Mathematical Functions The experiments were implemented in Python. Each objective function was evaluated using 20 independent random seeds for all algorithms to ensure statistical reliability.

The benchmark functions were tested under 30-dimensional (30D) and 1000-dimensional (1000D) settings to evaluate scalability across medium- and high-dimensional optimization problems.

The following settings were used across all algorithms:

- Population size: 30
- Number of iterations: 500

For HPPSO, the acceleration coefficients were set to $C_1 = 2$ and $C_2 = 2$. The maximum particle velocity was limited to

$$0.2 \times (\text{varMax} - \text{varMin}),$$

where **varMin** and **varMax** denote the lower and upper search-space boundaries. This velocity bounding scheme is widely used in PSO variants [Engelbrecht et al., 2007, p. 304]. When a particle violates the search boundaries, its velocity vector is reflected by sign inversion as recommended in [Padhye et al., 2012, p. 293].

The remaining HPPSO hyperparameters were set as follows:

$$\eta = 0.01, \quad \theta = 0.5, \quad \lambda = 0.4, \quad T_s = 10.$$

For standard PSO, the inertia weight was fixed at $w = 0.65$. The acceleration coefficients were set to $C_1 = C_2 = 1.5$ for PSO, PSO-M, and PSO-RIW. Higher nominal acceleration coefficients were assigned to HPPSO because conformity modulation scales the effective coefficients within the interval $[0, C_i]$.

For PSO-RIW, the inertia weight was sampled independently at each iteration from the uniform distribution

$$w \sim \mathcal{U}(0.4, 0.9).$$

For PSO-M, the mutation rate was set to 0.05 and the mutation scale to 0.1. Velocity bounding was applied only in HPPSO.

For SHADE, the history size parameter was set to $H = 100$.

For CSA, the parameter w was set equal to a , with the initial value of a approximately π , while parameter c was set to 0.8. The CSA algorithm is typically initialized using identical values across all dimensions, which may favor performance on simple benchmark functions but limit its effectiveness on more complex problems. In this study, we adopt a consistent initialization strategy across all compared algorithms, including CSA, to ensure a fair experimental comparison. This choice allows for a more uniform evaluation setting across different methods and benchmark problems. The results suggest that CSA performance is sensitive to initialization schemes, particularly in heterogeneous per-dimension settings, which may contribute to its reduced performance on the considered benchmark suite.

For CMAES, the initial step-size (σ) was set to one-fourth of the search space range. In the 1000-dimensional setting, SEP-CMAES was used instead of standard CMAES due to scalability limitations in high-dimensional optimization problems.

For GA-MPC, the parameters were set as follows:

$$cr = 1.0, \quad p = 0.1, \quad \mu = 0.7, \quad \sigma = 0.1,$$

with an archive ratio of 0.5. The tournament selection size was randomly chosen between 2 and 3.

Additional implementation details and source code are provided in the supplementary material.

Results on Mathematical Functions

Convergence Curves Figures 7 (a)–(o) and Figure 8 (a)–(o) present the median convergence curves for selected benchmark functions under 30-dimensional and 1000-dimensional settings, respectively.

The results indicate that standard PSO variants, including PSO, PSO-RIW, and PSO-M, exhibit premature convergence across most benchmark functions. In contrast, HPPSO demonstrates a smoother convergence behavior and maintains exploration over a larger number of iterations within the fixed evaluation budget.

HPPSO shows competitive performance compared to state-of-the-art methods such as CMAES and SHADE, while outperforming GA-MPC and GWO in most test cases. However, performance on some multimodal functions like *Michalewicz*, *Ackley* and *Levy* remains limited, and HPPSO improvement rate though not converged yet is slow and this suggests the need to incorporate some operator or solution that may increase the improvement rate of the algorithm. In several cases, the convergence curves suggest that HPPSO has not yet fully stabilized within the given iteration budget, which may indicate potential for further improvement under extended evaluations.

A similar trend is observed in the 1000-dimensional setting, suggesting that HPPSO maintains stable behavior across different problem dimensionalities. Within the fixed iteration budget, SHADE exhibits noticeable performance degradation compared to HPPSO, while HPPSO achieves the second-best overall ranking after SEP-CMAES. SEP-CMAES, together with its standard CMAES variant, remains one of the most competitive approaches for high-dimensional continuous optimization.

An additional observation is that GWO demonstrates improved relative performance in the 1000-dimensional setting. This behavior contrasts with some recent studies that criticized it, indicating that further investigation into its efficiency in extremely high-dimensional optimization problems may be worthwhile.

HPPSO outperforms both SEP-CMAES and CMAES primarily on the *Discus* function, which is characterized by severe ill-conditioning induced by highly anisotropic scaling. This observation may indicate that HPPSO exhibits favorable coordinate-wise search behavior in strongly stretched yet separable landscapes.

In contrast, the baseline PSO variants exhibit the weakest overall performance, while GA-MPC demonstrates moderate competitiveness across the benchmark suite.

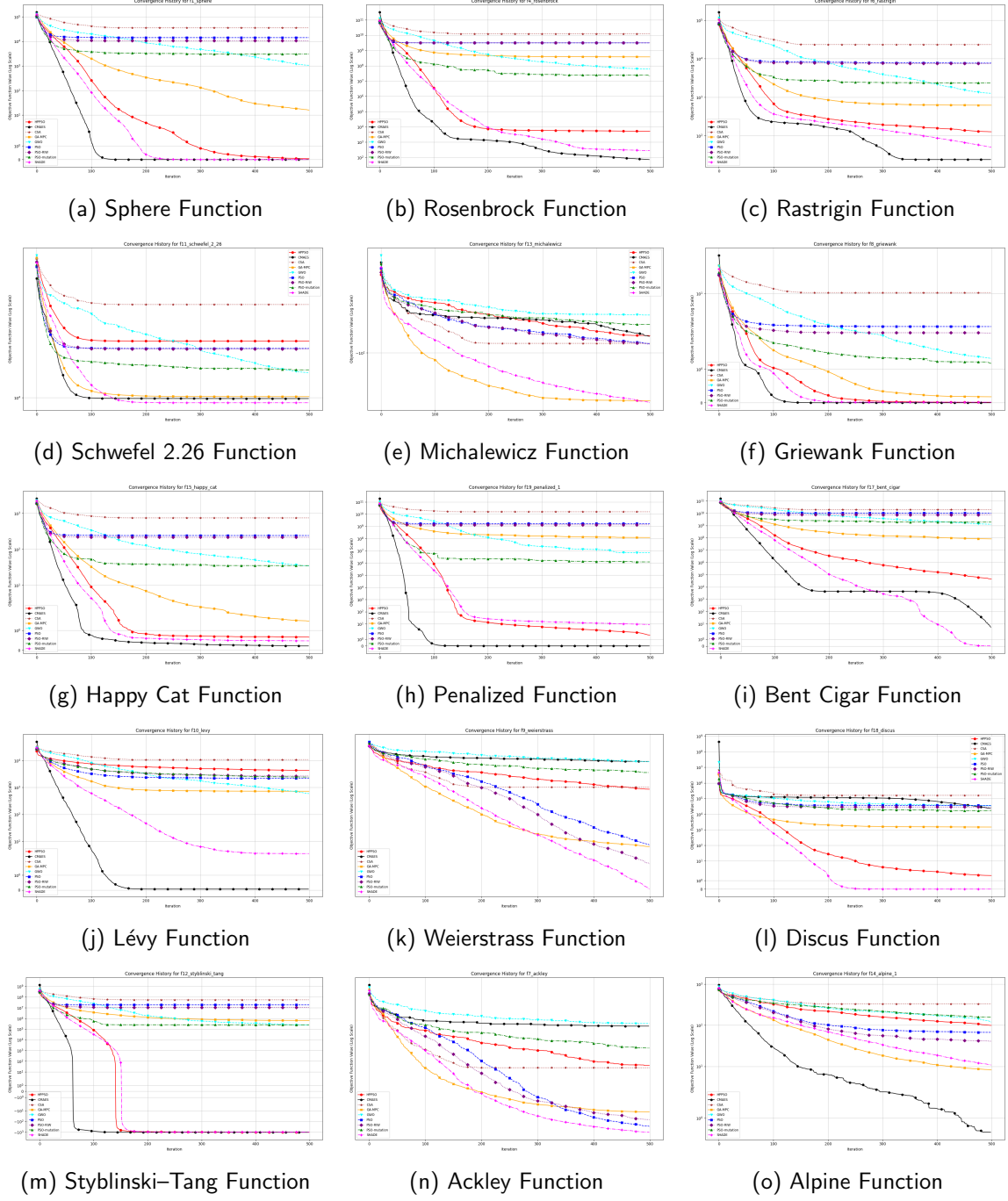
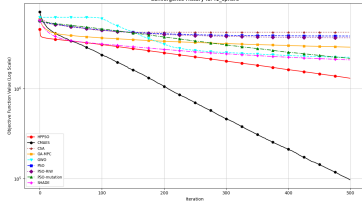
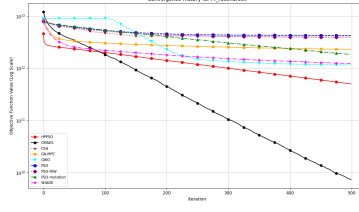


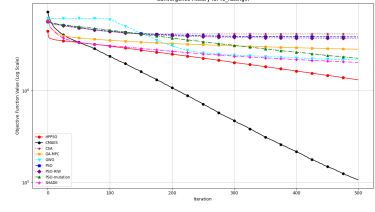
Figure 7: Median convergence curves of HPPSO on 30-dimensional benchmark functions over 20 independent runs. All functions are evaluated under shifted problem instances.



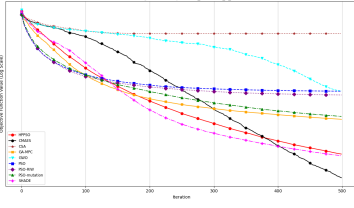
(a) Sphere Function



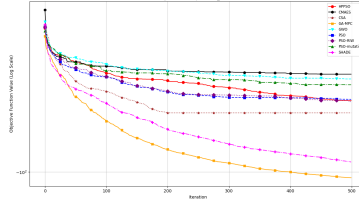
(b) Rosenbrock Function



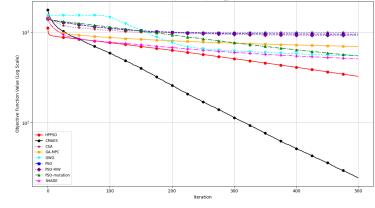
(c) Rastrigin Function



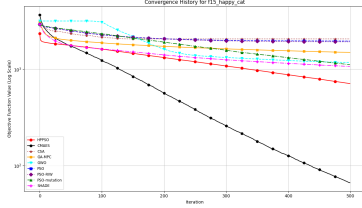
(d) Schwefel 2.26 Function



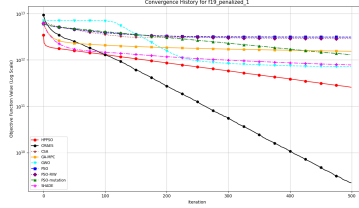
(e) Michalewicz Function



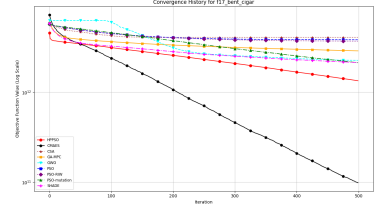
(f) Griewank Function



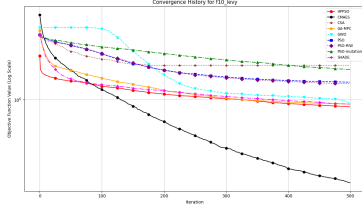
(g) Happy Cat Function



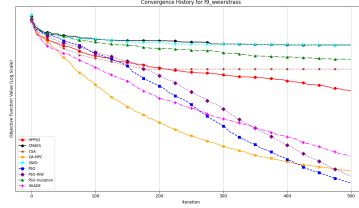
(h) Penalized Function



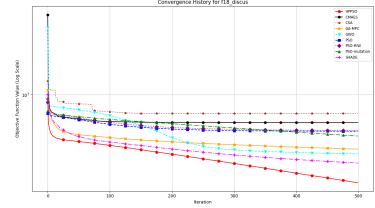
(i) Bent Cigar Function



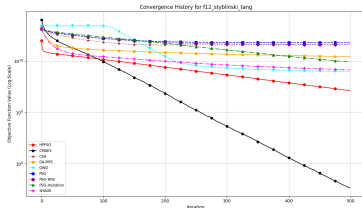
(j) Lévy Function



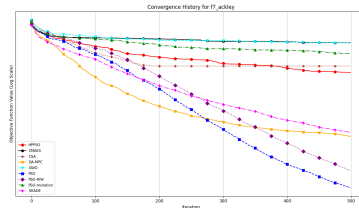
(k) Weierstrass Function



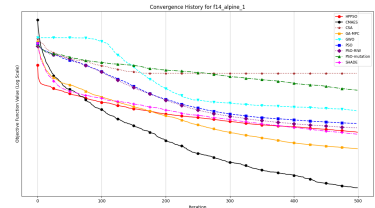
(l) Discus Function



(m) Styblinski–Tang Function



(n) Ackley Function



(o) Alpine Function

Figure 8: Median convergence curves of HPPSO on 1000-dimensional benchmark functions over 20 independent runs. All functions are evaluated under shifted problem instances.

Numerical Analysis of Results on Mathematical Functions (30D and 1000D) Tables 2 and 3 present the statistical results of all algorithms on the shifted benchmark functions in 30-dimensional and 1000-dimensional settings, respectively. The reported values correspond to the mean and standard deviation over 20 independent runs. Algorithm rankings are computed based on the lowest mean performance for each test function.

The results in Table 4 indicate that SHADE achieves the best overall ranking in the 30D setting, followed by CMAES and GA-MPC. HPPSO ranks fourth in this setting.

In contrast, Table 5 shows that HPPSO achieves the best average rank in the 1000D setting, followed closely by SHADE and CMAES. Although HPPSO attains the top average ranking, the differences between the leading algorithms are relatively small (HPPSO ≈ 2.93 , SHADE ≈ 3.10 , CMAES ≈ 3.25), indicating competitive performance among these methods in high-dimensional scenarios.

To assess whether the observed differences are statistically significant, non-parametric tests such as the Wilcoxon signed-rank test or Friedman test can be applied to compare algorithm performance across multiple functions. To evaluate the comparative performance of the considered optimization algorithms, we employ pairwise statistical comparisons based on the Wilcoxon signed-rank test. The results of these comparisons are summarized in Tables 6 and 7. In addition, a simple aggregated scoring metric is computed as:

$$\text{Score} = (\text{number of wins} + 0.5 \times \text{number of ties}) \times 100,$$

which provides a global ranking perspective across all benchmark functions.

In the 30-dimensional setting, CMAES and SHADE achieve identical scores, placing them at the top of the ranking. HPPSO follows as the next best-performing method, outperforming classical PSO variants, which consistently rank lower across the benchmark suite. This indicates that HPPSO remains competitive with state-of-the-art optimizers in moderate-dimensional problems.

For the 1000-dimensional case, SEP-CMAES obtains the highest score, while HPPSO ranks second overall. This result suggests that HPPSO maintains strong performance in high-dimensional settings and remains competitive with leading evolutionary strategies. In contrast, several baseline methods, including PSO variants, show degraded performance as dimensionality increases, highlighting the scalability advantages of HPPSO.

Function	PSO	PSO-M	PSO-RIW	HPPSO	GA-MPC	GWO	SHADE	CSA	CMAES
f_1	1.41E+04 9.70E+03 8	3.02E+03 1.63E+03 6	1.06E+04 8.75E+03 7	6.42E-02 4.99E-02 3	1.59E+01 4.07E+01 4	1.06E+03 8.91E+02 5	5.95E-09 1.30E-08 2	3.54E+04 1.49E+04 9	8.73E-11 1.14E-11 1
f_2	9.10E+02 3.19E+02 4	3.59E+19 1.57E+20 7	1.01E+03 2.25E+02 5	1.09E+03 3.14E+02 6	2.92E+01 5.96E+01 3	1.02E+22 4.46E+22 8	2.99E-02 4.98E-02 2	2.05E+38 8.96E+38 9	2.76E-06 8.78E-06 1
f_3	8.07E+04 2.33E+04 5	7.42E+04 1.66E+04 4	8.25E+04 3.42E+04 6	2.19E+04 9.01E+03 2	3.86E+04 1.19E+04 3	9.26E+04 1.86E+04 7	1.06E+04 8.84E+03 1	5.27E+05 1.46E+06 9	1.17E+05 2.30E+04 8
f_4	3.13E+09 3.61E+09 7	2.14E+07 3.17E+07 4	3.15E+09 2.92E+09 8	5.21E+03 5.70E+03 3	3.79E+08 1.10E+09 6	6.22E+07 7.91E+07 5	2.80E+02 4.33E+02 2	1.22E+10 1.37E+10 9	2.79E+02 3.64E+02 1
f_5	8.45E+01 2.58E+01 6	8.94E+01 2.81E+01 7	1.14E+02 3.31E+01 8	3.01E+01 7.17E+00 4	6.07E+00 6.23E+00 1	5.70E+01 9.96E+00 5	1.10E+01 4.68E+00 2	1.92E+02 6.32E+01 9	1.80E+01 6.05E+00 3
f_6	7.69E+03 4.48E+03 8	2.36E+03 1.56E+03 6	7.50E+03 6.27E+03 7	1.26E+02 2.86E+01 3	6.18E+02 7.24E+02 4	1.26E+03 5.78E+02 5	5.04E+01 1.84E+01 2	2.33E+04 1.46E+04 9	2.56E+01 6.39E+00 1
f_7	2.01E+01 1.19E-01 2	2.08E+01 1.49E-01 7	2.02E+01 1.84E-01 3	2.07E+01 1.59E-01 6	2.02E+01 1.30E-01 4	2.11E+01 7.57E-02 8	2.01E+01 1.18E-02 1	2.07E+01 1.81E-01 5	2.11E+01 4.52E-02 9
f_8	2.83E+00 1.00E+00 8	1.17E+00 6.77E-01 5	2.21E+00 1.23E+00 7	1.50E-02 1.35E-02 2	1.79E-01 4.17E-01 4	1.32E+00 3.17E-01 6	1.90E-02 1.58E-02 3	1.03E+01 4.52E+00 9	8.81E-11 1.03E-11 1
f_9	1.98E+01 7.09E+00 4	3.85E+01 1.95E+00 7	1.67E+01 5.79E+00 2	3.30E+01 2.89E+00 5	1.95E+01 4.01E+00 3	4.25E+01 1.40E+00 9	1.32E+01 1.40E+00 1	3.36E+01 4.37E+00 6	4.25E+01 1.31E+00 8
f_{10}	2.20E+03 1.00E+03 5	2.42E+03 1.44E+03 6	2.65E+03 1.15E+03 7	4.31E+03 1.50E+03 8	7.20E+02 6.69E+02 4	5.93E+02 3.22E+02 3	3.51E+00 3.31E+00 2	1.08E+04 4.27E+03 9	2.27E-02 9.90E-02 1
f_{11}	1.06E+04 1.44E+02 7	1.04E+04 1.91E+02 5	1.06E+04 2.18E+02 6	1.07E+04 2.54E+02 8	1.00E+04 1.20E+02 2	1.03E+04 1.51E+02 4	9.94E+03 1.49E+02 1	1.12E+04 2.96E+02 9	1.03E+04 1.23E+02 3
f_{12}	1.82E+07 1.98E+07 8	2.50E+05 2.35E+05 5	1.08E+07 9.33E+06 7	-1.04E+03 3.83E+01 1	6.19E+05 8.49E+05 6	2.39E+05 2.31E+05 4	-9.88E+02 3.15E+01 2	5.44E+07 6.52E+07 9	-2.58E+02 1.26E+02 3
f_{13}	-8.82E+00 1.90E+00 4	-6.83E+00 9.59E-01 7	-8.89E+00 1.48E+00 3	-7.93E+00 1.60E+00 6	-1.90E+01 1.86E+00 2	-6.02E+00 5.96E-01 8	-1.95E+01 1.05E+00 1	-8.80E+00 2.22E+00 5	-2.54E+00 7.10E-01 9
f_{14}	6.71E+01 4.10E+01 5	1.57E+02 7.65E+01 8	4.09E+01 3.48E+01 4	9.93E+01 3.21E+01 6	8.22E+00 7.51E+00 2	1.19E+02 6.10E+01 7	1.08E+01 4.25E+00 3	3.31E+02 7.73E+01 9	3.23E-06 1.28E-05 1
f_{15}	2.43E+02 1.32E+02 8	3.53E+01 2.48E+01 6	2.16E+02 1.62E+02 7	6.58E-01 1.40E-01 3	1.47E+00 1.51E+00 4	3.46E+01 1.59E+01 5	4.70E-01 1.25E-01 2	7.52E+02 4.60E+02 9	2.69E-01 5.89E-02 1
f_{16}	9.28E+03 5.92E+03 8	1.61E+03 1.12E+03 6	5.30E+03 5.27E+03 7	6.53E-01 2.44E-01 3	2.88E+02 5.91E+02 4	1.33E+03 8.38E+02 5	5.58E-01 2.81E-01 2	2.78E+04 1.32E+04 9	4.58E-01 6.46E-02 1
f_{17}	1.02E+10 6.65E+09 8	1.90E+09 1.83E+09 6	7.49E+09 6.76E+09 7	4.48E+04 3.21E+04 3	8.17E+07 2.73E+08 4	1.27E+09 8.45E+08 5	2.26E-02 3.84E-02 2	1.96E+10 1.30E+10 9	2.23E-08 2.39E-08 1
f_{18}	3.62E+04 2.67E+04 8	1.67E+04 1.77E+04 5	2.85E+04 2.13E+04 6	1.57E+00 2.92E+00 3	1.49E+03 3.22E+03 4	3.42E+04 1.20E+04 7	3.29E-08 4.82E-08 2	1.61E+05 8.90E+04 9	8.85E-11 1.27E-11 1
f_{19}	1.70E+09 1.81E+09 8	1.22E+06 2.13E+06 4	1.34E+09 1.98E+09 7	1.61E+00 2.10E+00 2	1.22E+08 3.08E+08 6	6.85E+06 1.04E+07 5	8.76E+00 4.88E+00 3	1.58E+10 1.95E+10 9	8.57E-11 1.26E-11 1
f_{20}	4.97E-05 8.80E-05 4	4.69E-04 6.92E-04 7	3.64E-05 6.61E-05 3	5.32E-05 9.45E-05 5	4.03E-08 6.11E-08 2	4.28E-04 2.54E-04 6	5.41E-10 1.13E-10 1	4.33E-03 1.27E-02 9	7.78E-04 7.21E-04 8

Table 2: Performance evaluation on shifted optimization benchmark functions in a 30-dimensional setting, reporting the mean and standard deviation of the objective function value over 20 independent runs. Methods are ranked in ascending order of mean objective value, where lower values indicate better performance.

Function	PSO	PSO-m	PSO-RIW	HPPSO	GA-MPC	GWO	SHADE	CSA	sep-CMAES
f_1	3.84E+06 1.49E+05 8	2.16E+06 8.79E+04 4	3.66E+06 2.03E+05 7	1.30E+06 9.88E+04 2	2.88E+06 1.05E+05 6	2.23E+06 3.33E+04 5	2.08E+06 9.50E+04 3	4.23E+06 4.41E+05 9	9.67E+04 6.86E+03 1
f_2	∞ 1	∞ 1	∞ 1	∞ 1	∞ 1	∞ 1	∞ 1	∞ 1	∞ 1
f_3	7.43E+06 8.98E+05 4	7.64E+06 7.94E+05 6	7.48E+06 8.57E+05 5	5.06E+06 4.07E+05 2	1.05E+07 3.60E+06 9	6.87E+06 5.32E+05 3	4.70E+06 6.21E+05 1	1.00E+07 2.13E+06 8	7.65E+06 6.12E+06 7
f_4	4.27E+12 3.36E+11 9	1.84E+12 1.51E+11 5	3.92E+12 5.14E+11 7	5.02E+11 5.06E+10 2	2.30E+12 1.89E+11 6	1.16E+12 3.35E+10 3	1.23E+12 1.00E+11 4	4.16E+12 1.30E+12 8	7.29E+09 7.71E+08 1
f_5	9.85E+03 2.76E+02 8	8.47E+03 2.17E+02 6	9.83E+03 4.05E+02 7	7.21E+03 3.12E+02 3	8.01E+03 3.17E+02 4	8.19E+03 1.76E+02 5	7.06E+03 2.75E+02 2	1.23E+04 5.54E+02 9	4.49E+03 6.30E+01 1
f_6	3.82E+06 1.39E+05 8	2.24E+06 7.31E+04 5	3.68E+06 1.82E+05 7	1.30E+06 9.86E+04 2	2.80E+06 1.15E+05 6	2.15E+06 4.64E+04 4	2.02E+06 4.91E+04 3	4.11E+06 4.59E+05 9	1.07E+05 4.48E+03 1
f_7	2.13E+01 6.28E-02 1	2.16E+01 1.83E-02 7	2.13E+01 9.15E-02 2	2.16E+01 3.10E-02 5	2.14E+01 2.36E-02 3	2.16E+01 5.80E-03 8	2.14E+01 8.22E-03 4	2.16E+01 3.46E-02 6	2.16E+01 7.88E-03 9
f_8	9.55E+02 3.33E+01 8	5.50E+02 2.24E+01 4	9.25E+02 5.57E+01 7	3.27E+02 2.80E+01 2	6.98E+02 2.85E+01 6	5.52E+02 9.02E+00 5	5.09E+02 2.45E+01 3	1.00E+03 1.06E+02 9	2.45E+01 1.04E+00 1
f_9	1.66E+03 4.65E+01 1	1.87E+03 1.72E+01 7	1.68E+03 4.92E+01 2	1.82E+03 4.16E+01 5	1.68E+03 2.78E+01 3	1.89E+03 6.35E+00 8	1.71E+03 8.82E+00 4	1.85E+03 2.59E+01 6	1.90E+03 8.92E+00 9
f_{10}	1.22E+06 8.01E+04 7	1.40E+06 8.98E+04 8	1.20E+06 7.64E+04 6	9.22E+05 5.79E+04 2	9.50E+05 4.53E+04 3	9.59E+05 6.99E+04 5	9.51E+05 4.03E+04 4	1.47E+06 1.92E+05 9	3.93E+05 3.67E+04 1
f_{11}	3.94E+05 1.60E+03 7	3.87E+05 2.43E+03 5	3.93E+05 1.86E+03 6	3.78E+05 1.24E+03 3	3.87E+05 2.10E+03 4	3.94E+05 3.64E+03 8	3.77E+05 2.09E+03 2	4.10E+05 2.58E+03 9	3.72E+05 1.28E+03 1
f_{12}	2.28E+10 1.42E+09 9	9.51E+09 7.22E+08 5	2.09E+10 2.71E+09 8	2.64E+09 3.14E+08 2	1.22E+10 1.07E+09 6	6.29E+09 2.08E+08 3	6.69E+09 4.93E+08 4	2.07E+10 4.98E+09 7	3.36E+07 3.68E+06 1
f_{13}	-4.31E+01 4.82E+00 5	-3.37E+01 2.38E+00 7	-4.09E+01 5.20E+00 6	-4.44E+01 1.43E+01 4	-1.08E+02 8.47E+00 1	-3.19E+01 1.45E+00 8	-8.78E+01 1.85E+00 2	-4.49E+01 9.72E+00 3	-1.54E+01 4.33E+00 9
f_{14}	2.74E+04 1.05E+03 6	3.19E+04 1.49E+03 8	2.68E+04 7.79E+02 5	2.63E+04 9.94E+02 4	2.44E+04 7.01E+02 2	2.90E+04 1.51E+03 7	2.61E+04 6.33E+02 3	3.45E+04 1.80E+03 9	2.04E+04 9.09E+02 1
f_{15}	1.97E+03 7.86E+01 8	1.12E+03 5.22E+01 4	1.93E+03 1.55E+02 7	7.11E+02 6.00E+01 2	1.50E+03 4.84E+01 6	1.18E+03 2.46E+01 5	1.07E+03 3.79E+01 3	2.07E+03 1.58E+02 9	6.63E+01 2.96E+00 1
f_{16}	3.86E+06 1.54E+05 8	2.16E+06 9.00E+04 4	3.70E+06 1.91E+05 7	1.29E+06 9.32E+04 2	2.85E+06 1.12E+05 6	2.25E+06 4.78E+04 5	2.07E+06 1.10E+05 3	4.23E+06 4.23E+05 9	9.67E+04 6.36E+03 1
f_{17}	3.85E+12 1.70E+11 8	2.14E+12 8.91E+10 3	3.72E+12 3.00E+11 7	1.35E+12 8.72E+10 2	2.90E+12 1.48E+11 6	2.27E+12 5.68E+10 5	2.15E+12 7.63E+10 4	4.09E+12 2.74E+11 9	9.85E+10 4.11E+09 1
f_{18}	4.75E+06 2.98E+05 6	4.31E+06 2.59E+05 5	4.83E+06 3.98E+05 7	1.66E+06 1.17E+05 1	3.30E+06 1.94E+05 4	2.99E+06 5.56E+04 3	2.48E+06 1.21E+05 2	6.83E+06 1.91E+06 9	5.67E+06 1.36E+05 8
f_{19}	3.12E+12 3.10E+11 9	1.29E+12 1.32E+11 5	2.88E+12 3.35E+11 7	2.58E+11 2.91E+10 2	1.54E+12 1.20E+11 6	7.19E+11 2.14E+10 3	7.80E+11 4.76E+10 4	3.08E+12 1.02E+12 8	2.24E+09 3.00E+08 1
f_{20}	1.00E-37 2.82E-37 3	2.19E-30 9.15E-30 6	5.68E-34 2.40E-33 4	1.15E-29 4.93E-29 8	8.61E-87 2.92E-86 2	6.39E-30 2.62E-29 7	1.08E-99 4.05E-99 1	1.05E-31 4.59E-31 5	1.84E-11 3.05E-11 9

Table 3: Performance evaluation on shifted optimization benchmark functions in a 1000-dimensional setting, reporting the mean and standard deviation of the objective function value over 20 independent runs. Methods are ranked in ascending order of mean objective value, where lower values indicate better performance.

Table 4: Average ranking of optimization algorithms across 20 shifted 30-dimensional benchmark functions, where rankings are computed based on performance over 20 independent runs per function.

Algorithm	Average Rank
SHADE	1.9200
CMAES	2.9500
GA-MPC	3.4650
HPPSO	4.1150
PSO-M	5.9650
PSO-RIW	6.0575
GWO	6.1450
PSO	6.2450
CSA	8.1375

Table 5: Average ranking of optimization algorithms across 20 shifted 1000-dimensional benchmark functions, where rankings are computed based on performance over 20 independent runs per function.

Algorithm	Average Rank
HPPSO	2.9250
SHADE	3.0975
sep-CMAES	3.2450
GA-MPC	4.6925
GWO	5.1725
PSO-M	5.4775
PSO-RIW	6.2225
PSO	6.4850
CSA	7.6825

Table 6: Statistical comparison of optimization algorithms based on the Wilcoxon signed-rank test across 20 shifted 30-dimensional benchmark functions, each evaluated over 20 independent runs.

Algorithm	Total Wins	Total Ties	Normalized Score (0–100)
CMAES	7	1	93.75
SHADE	7	1	93.75
HPPSO	6	0	75.00
GA-MPC	5	0	62.00
PSO-RIW	4	0	50.00
PSO	3	0	37.50
GWO	1	1	18.75
PSO-M	1	1	18.75
CSA	0	0	0.00

Table 7: Statistical comparison of optimization algorithms based on the Wilcoxon signed-rank test across 20 shifted 1000-dimensional benchmark functions, each evaluated over 20 independent runs.

Algorithm	Total Wins	Total Ties	Normalized Score (0–100)
SEP-CMAES	8	0	100.00
HPPSO	7	0	87.5
GWO	6	0	75.0
SHADE	5	0	62.5
PSO-M	4	0	50.0
GA-MPC	3	0	37.5
PSO-RIW	2	0	25.0
PSO	1	0	12.5
CSA	0	0	0.00

6.2.2 CEC 2011 Benchmark Problems

To further evaluate the effectiveness of the proposed algorithm, experiments were conducted on the CEC 2011 benchmark suite. These benchmark problems are widely regarded as challenging real-world optimization tasks and contain heterogeneous search characteristics that make them suitable for evaluating optimization robustness and scalability.

The experiments were conducted using the `minionpy` library. For consistency, the same hyperparameter settings were used for all algorithms. The population size was fixed at 200, while the maximum number of iterations was set to 1000. In addition, 10 independent random seeds were used for each benchmark problem.

It should be noted that CMAES was excluded from benchmark problems with one-dimensional search spaces, since the algorithm requires problem dimensionality greater than one ($d > 1$).

Table 8 presents the mean and standard deviation obtained by each algorithm across the considered benchmark problems, together with the corresponding performance rankings.

Table 8: Performance evaluation of optimization algorithms on real-world CEC2011 benchmark problems. The table reports the mean and standard deviation of objective values over 10 independent runs for each problem. Algorithms are ranked in ascending order of mean objective value (lower is better).

Problem	Dim	Metric	CMAES	CSA	GA-MPC	GWO	HPPSO	PSO	PSO-RIW	PSO-M	SHADE
T1	6	Mean Std Rank	1.38E+01 2.03E+00 7	2.09E+01 5.26E+00 9	9.29E+00 6.42E+00 2	1.27E+01 7.27E+00 4	1.81E+01 6.44E+00 8	1.29E+01 7.43E+00 5	1.35E+01 6.25E+00 6	1.19E+01 6.33E+00 3	8.31E-02 1.32E-01 1
T2	30	Mean Std Rank	-1.71E+00 3.83E-01 9	-8.62E+00 3.26E+00 8	-1.35E+01 4.34E+00 7	-2.38E+01 2.69E+00 3	-2.67E+01 6.25E-01 1	-1.62E+01 4.41E+00 4	-1.56E+01 3.42E+00 6	-1.56E+01 3.90E+00 5	-2.41E+01 5.90E-01 2
T3	1	Mean Std Rank	— — —	1.15E-05 3.12E-19 5	1.15E-05 5.42E-19 7	1.15E-05 4.28E-16 8	1.15E-05 1.14E-19 4	1.15E-05 1.43E-19 2	1.15E-05 1.54E-19 3	1.15E-05 2.97E-19 6	1.15E-05 1.32E-19 1
T4	1	Mean Std Rank	— — —	1.34E+02 4.92E-14 5	1.34E+02 3.11E-14 7	1.34E+02 6.27E-12 8	1.34E+02 2.70E-14 3	1.34E+02 4.58E-14 1	1.34E+02 2.98E-14 3	1.34E+02 4.12E-14 6	1.34E+02 3.48E-14 1
T5	30	Mean Std Rank	-1.43E+01 1.46E+00 9	-2.63E+01 4.52E+00 7	-2.18E+01 1.31E+00 8	-3.31E+01 1.18E+00 3	-3.33E+01 2.62E+00 2	-3.16E+01 4.40E+00 4	-3.15E+01 2.90E+00 5	-3.12E+01 3.13E+00 6	-3.51E+01 4.95E-01 1
T6	30	Mean Std Rank	-3.09E+00 1.61E+00 9	-1.71E+01 4.15E+00 7	-1.67E+01 1.36E+00 8	-2.45E+01 3.37E+00 2	-1.87E+01 2.29E+00 6	-2.04E+01 2.36E+00 5	-2.09E+01 3.46E+00 3	-2.06E+01 1.93E+00 4	-2.90E+01 1.03E-01 1
T7	20	Mean Std Rank	5.71E-01 7.15E-02 1	1.37E+00 1.86E-01 8	1.79E+00 1.03E-01 9	1.15E+00 3.14E-01 6	1.11E+00 1.07E-01 5	1.04E+00 1.62E-01 2	1.04E+00 8.27E-02 3	1.10E+00 1.64E-01 4	1.22E+00 7.68E-02 7
T8	7	Mean Std Rank	2.51E+02 2.05E+01 8	3.58E+02 1.51E+02 9	2.20E+02 0.00E+00 1	2.20E+02 0.00E+00 1	2.33E+02 2.31E+01 7	2.23E+02 9.30E+00 5	2.20E+02 0.00E+00 1	2.23E+02 9.30E+00 5	2.20E+02 0.00E+00 1
T9	126	Mean Std Rank	2.53E+03 4.74E+02 2	1.11E+06 3.38E+05 9	4.08E+04 5.52E+03 5	1.33E+04 6.27E+03 4	2.76E+03 9.84E+02 3	2.86E+05 1.77E+05 7	2.90E+05 1.31E+05 8	1.75E+05 1.10E+05 6	2.18E+03 3.67E+02 1
T10	12	Mean Std Rank	-6.80E-01 4.10E-01 9	-5.54E+00 1.15E+00 8	-8.18E+00 4.27E-02 2	-8.04E+00 1.22E-01 3	-7.87E+00 1.63E-01 4	-7.37E+00 3.11E-01 5	-7.19E+00 4.90E-01 6	-6.97E+00 7.81E-01 7	-8.20E+00 1.64E-02 1
T11	120	Mean Std Rank	5.21E+04 4.22E+02 1	6.60E+07 4.22E+07 9	1.05E+06 2.42E+05 4	1.16E+06 7.24E+05 5	5.23E+04 5.15E+02 3	2.82E+06 4.96E+06 7	1.68E+06 3.93E+06 6	4.77E+06 5.45E+06 8	5.22E+04 3.70E+02 2
T12	240	Mean Std Rank	1.07E+06 1.23E+03 1	7.48E+06 1.25E+06 9	1.20E+06 1.98E+04 5	1.17E+06 6.03E+04 4	1.08E+06 2.37E+03 2	1.86E+06 4.08E+05 7	1.59E+06 2.77E+05 6	2.00E+06 3.50E+05 8	1.08E+06 1.32E+03 3
T13	6	Mean Std Rank	1.55E+04 2.48E+01 5	1.55E+04 6.67E+01 9	1.54E+04 2.85E-05 2	1.55E+04 2.09E+01 4	1.55E+04 2.72E+01 3	1.55E+04 3.34E+01 8	1.55E+04 2.15E+01 7	1.55E+04 2.47E+01 6	1.54E+04 8.49E-07 1
T14	13	Mean Std Rank	1.81E+04 2.33E+01 1	1.92E+04 1.60E+02 8	1.87E+04 5.14E+01 3	1.91E+04 1.29E+02 7	1.89E+04 2.20E+02 4	1.92E+04 1.48E+02 9	1.91E+04 1.37E+02 5	1.91E+04 2.01E+02 6	1.81E+04 6.80E+01 2
T15	15	Mean Std Rank	3.27E+04 2.73E+01 1	4.56E+04 3.16E+04 9	3.28E+04 4.25E+01 3	3.30E+04 1.28E+02 8	3.30E+04 1.06E+02 6	3.30E+04 1.23E+02 5	3.30E+04 1.05E+02 7	3.29E+04 8.41E+01 4	3.27E+04 6.77E+01 2
T16	40	Mean Std Rank	1.24E+05 5.62E+02 1	1.48E+05 1.03E+04 9	1.36E+05 3.46E+03 3	1.38E+05 4.32E+03 4	1.38E+05 2.57E+03 5	1.42E+05 5.85E+03 7	1.44E+05 4.26E+03 8	1.40E+05 5.82E+03 6	1.28E+05 3.02E+03 2
T17	140	Mean Std Rank	1.94E+06 1.12E+04 2	4.33E+09 2.86E+09 9	2.99E+06 6.29E+05 6	2.45E+06 7.94E+05 5	1.96E+06 2.00E+04 4	1.95E+06 2.70E+04 3	1.91E+07 2.95E+07 8	6.00E+06 1.20E+07 7	1.87E+06 1.85E+04 1
T18	96	Mean Std Rank	9.42E+05 1.93E+03 2	7.61E+06 7.35E+06 9	2.34E+06 4.53E+05 8	9.83E+05 9.87E+03 3	1.02E+06 2.11E+05 4	1.41E+06 6.95E+05 6	1.40E+06 6.52E+05 5	1.43E+06 6.67E+05 7	9.35E+05 2.17E+03 1
T19	96	Mean Std Rank	9.43E+05 2.23E+03 2	7.99E+06 7.06E+06 9	3.02E+06 4.39E+05 8	1.35E+06 1.68E+05 3	1.38E+06 2.21E+05 4	1.65E+06 1.13E+06 7	1.48E+06 6.43E+05 5	1.52E+06 5.63E+05 6	9.42E+05 1.21E+03 1
T20	96	Mean Std Rank	9.53E+05 2.67E+03 2	7.62E+06 7.35E+06 9	2.58E+06 3.66E+05 8	9.95E+05 8.04E+03 3	1.01E+06 1.20E+05 4	1.42E+06 6.96E+05 6	1.41E+06 6.52E+05 5	1.43E+06 6.72E+05 7	9.44E+05 2.10E+03 1
T21	26	Mean Std Rank	2.99E+01 3.28E+00 2	4.42E+01 7.67E+00 9	2.77E+01 3.32E+00 1	3.60E+01 3.86E+00 7	3.23E+01 3.73E+00 4	3.64E+01 3.10E+00 8	3.48E+01 2.02E+00 5	3.60E+01 5.93E+00 6	3.19E+01 1.59E+00 3
T22	22	Mean Std Rank	2.40E+01 7.68E+00 8	3.00E+01 9.75E+00 9	1.44E+01 1.52E+00 2	1.91E+01 1.84E+00 4	1.79E+01 2.78E+00 3	2.15E+01 4.15E+00 6	2.24E+01 3.55E+00 7	2.01E+01 3.05E+00 5	1.26E+01 7.66E-01 1

To summarize the overall performance across all CEC 2011 benchmark problems, the average rank of each algorithm is reported in Table 9. The results indicate that SHADE achieves the best overall ranking, demonstrating superior performance on these challenging real-world optimization tasks. HPPSO ranks second, followed closely by CMAES, indicating that HPPSO remains highly effective for heterogeneous real-world optimization problems.

Table 9: Average performance ranking of optimization algorithms on the CEC2011 benchmark suite, based on results obtained from 10 independent runs for each problem.

Algorithm	Average Rank
SHADE	1.68
HPPSO	4.05
CMAES	4.10
GWO	4.50
GA-MPC	4.95
PSO-RIW	5.36
PSO	5.41
PSO-m	5.82
CSA	8.27

Interestingly, HPPSO outperforms GA-MPC, the winner of the original CEC 2011 competition. Specifically, HPPSO achieves an average rank of 4.05, whereas GA-MPC obtains an average rank of 4.95. This result highlights the robustness and competitiveness of the proposed approach relative to established state-of-the-art optimization algorithms.

Table 10: Statistical comparison of optimization algorithms using the Wilcoxon signed-rank test across 22 CEC2011 benchmark problems, each evaluated over 10 independent runs.

Algorithm	Total Wins	Total Ties	Normalized Score (0–100)
SHADE	8	0	100.00
CMAES	5	2	75.00
HPPSO	4	3	68.75
GA-MPC	1	6	50.00
PSO	1	5	43.75
GWO	1	4	37.50
PSO-M	1	4	37.50
PSO-RIW	1	4	37.50
CSA	0	0	0.00

To further validate these findings, a Wilcoxon signed-rank test was conducted to assess the statistical significance of pairwise performance differences between algorithms. The results are summarized in Table 10. The statistical analysis indicates that SHADE and CMAES demonstrate significant superiority over most competing methods, followed by HPPSO, which exhibits competi-

tive performance relative to other state-of-the-art metaheuristics. Furthermore, HPPSO achieves a higher overall ranking score than GA-MPC, further supporting its effectiveness on complex benchmark problems.

Classical PSO also demonstrates relatively strong performance on several CEC 2011 problems, particularly those with low dimensionality. For example, PSO performs favorably on problems T3 and T4, which are one-dimensional optimization tasks.

Overall, the CEC 2011 benchmark suite does not primarily consist of extremely high-dimensional optimization problems. Consequently, the advantages of HPPSO are not as pronounced in this benchmark set as they are in large-scale optimization scenarios. Nevertheless, HPPSO remains highly competitive and achieves performance comparable to several state-of-the-art metaheuristics. However, SHADE and CMAES still demonstrate superior overall performance on these problems.

One possible explanation is that HPPSO relies heavily on stochastic exploration mechanisms and incorporates multiple randomized operators. While these characteristics improve exploration capability and help the algorithm avoid premature convergence in extremely large-dimensional search spaces, they may reduce exploitation precision near the optimum. As a result, HPPSO appears particularly effective at identifying promising search regions, although it may be less efficient in fine-grained local refinement compared to algorithms such as CMAES and SHADE.

This observation suggests a promising future research direction. In particular, HPPSO could potentially benefit from the integration of an additional exploitation or local refinement stage designed to enhance final solution accuracy after the exploratory search phase.

6.2.3 Real-World Application (Neural Network Training)

In this section, we evaluate the proposed HPPSO algorithm on a real-world optimization task and compare its performance against several baseline metaheuristic algorithms. Since HPPSO is designed to address high-dimensional optimization problems effectively, neural network training was selected as the target application due to the large number of trainable parameters involved in such models.

To assess the optimization capability of the algorithms, we considered the task of training neural networks for both classification and regression problems. Specifically, four classification datasets and four regression datasets were used in the experiments. In this framework, the neural network weights and biases constitute the decision variables to be optimized by the metaheuristic algorithms.

Neural Network Architecture Fully connected feedforward neural networks with two hidden layers were employed for all experiments. The number of neurons in the first hidden layer was set to two-thirds of the number of input features for each dataset, while the second hidden layer contained two-thirds of the neurons used in the first hidden layer.

Each optimization algorithm was evaluated using five independent random seeds, and the median performance across these runs was reported to reduce the influence of stochastic variations.

Table 11 summarizes the characteristics of the datasets used in the experiments, including the task type, number of samples, and number of input features. Table 12 presents the dimensionality of the corresponding neural network optimization problems.

Table 11: Summary of Dataset Characteristics Including Task Type, Number of Samples, and Number of Features

Dataset	Task	Samples	Features
California Housing	Regression	20640	8
Diabetes	Regression	442	10
Linnerud	Regression	20	3
Synthetic Regression	Regression	500	13
Wine	Classification	178	13
Breast Cancer	Classification	569	30
Digits	Classification	1797	64
Synthetic Classification	Classification	500	13

Table 12: Dimensionality of Each Dataset Problem Using a Two-Hidden-Layer Neural Network

Dataset	Dimensionality
California Housing	93
Diabetes	126
Linnerud	17
Synthetic Regression	170
Wine	188
Breast Cancer	796
Digits	3203
Synthetic Classification	186

Results of Neural Network Training Table 13 presents the median minimum cost achieved by each optimization algorithm during neural network training across all datasets. Overall, sep-CMAES achieves the lowest cost on most low- and moderate-dimensional datasets, while HPPSO exhibits its strongest performance in high-dimensional cases, particularly for the Breast Cancer and Digits datasets.

In lower-dimensional problems, HPPSO performs comparably to CMAES. For instance, in the California Housing dataset, HPPSO achieves a median MSE of 0.313, while CMAES and SHADE both achieve 0.31. This indicates only a marginal difference among these methods in moderate-dimensional settings.

In contrast, HPPSO demonstrates clear advantages in high-dimensional optimization. In the Digits dataset, which has the highest dimensionality among the considered problems, HPPSO achieves a median MSE of 0.063, compared to 0.256 for CMAES and 0.099 for SHADE. This indicates a substantial performance gap in favor of HPPSO in this setting.

SHADE also shows consistently strong and stable performance across most datasets, confirming its robustness across varying dimensionalities. However, in the highest-dimensional problems, HPPSO slightly outperforms SHADE, although the difference remains relatively small (0.063 vs. 0.099 for Digits).

These results suggest that HPPSO is particularly effective in high-dimensional search spaces. This behavior can be attributed to its strong exploratory capability and stochastic search dynamics, which help mitigate premature convergence. In contrast, classical PSO variants are more prone to

premature convergence and may become trapped in local optima, particularly in high-dimensional landscapes.

Unlike traditional PSO variants, HPPSO maintains higher population diversity during the optimization process, which allows it to explore promising regions of the search space more effectively before exploitation.

Figure 9 illustrates the ranking of algorithms as a function of problem dimensionality based on median MSE performance. HPPSO consistently ranks among the top-performing methods, achieving particularly strong results in high-dimensional problems while remaining competitive in lower-dimensional settings.

CMAES is not applicable to certain high-dimensional cases due to its computational complexity of $O(n^2)$, while its separable variant does not always retain the same level of performance in very high-dimensional settings. SHADE remains a strong competitor; however, it requires an external historical memory component (H), whereas HPPSO does not.

Overall, HPPSO demonstrates competitive performance with state-of-the-art metaheuristics such as SHADE and CMAES, particularly in high-dimensional and heterogeneous optimization problems. Although this study focuses on static benchmarks, including mathematical functions, CEC 2011, and neural network training tasks, the results suggest that HPPSO may also be promising for dynamic optimization problems such as robotics path planning and time-varying environments. However, this remains speculative and requires further empirical validation in future work.

Table 13: Median training mean squared error (MSE) achieved by each optimization algorithm across all neural network training datasets, evaluated over five independent runs per dataset.

Dataset	HPPSO	PSO	PSO-m	PSO-RIW	GWO	CSA	sep-CMAES	SHADE	GA-MPC
BreastCancer	0.041	0.083	0.095	0.078	0.086	0.184	0.053	0.042	0.041
CaliforniaHousing	0.313	0.498	0.403	0.502	0.406	0.738	0.31	0.31	0.328
Diabetes	0.402	0.472	0.453	0.474	0.461	0.59	0.384	0.397	0.412
Digits	0.063	0.284	0.424	0.22	0.257	0.553	0.256	0.099	0.392
Linnerud	0.88	0.881	0.881	0.881	0.88	0.88	0.883	0.88	0.88
SyntheticClassification	0.239	0.437	0.412	0.438	0.41	0.548	0.162	0.208	0.257
SyntheticRegression	0.017	0.141	0.086	0.151	0.039	0.503	0.01	0.02	0.033
Wine	0.012	0.093	0.047	0.09	0.062	0.17	0.007	0.012	0.015

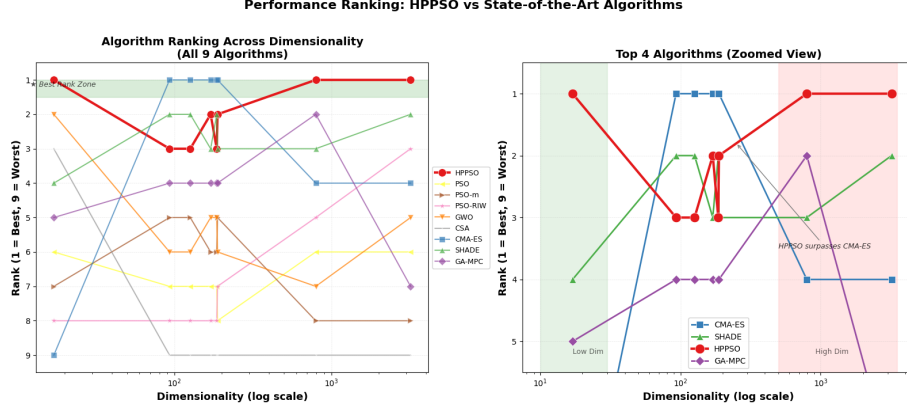


Figure 9: Ranking variation with respect to dataset dimensionality in neural network training. Rankings are based on the lowest training MSE over five independent runs per algorithm. The left axis shows all algorithms, while the right axis highlights the top four methods (HPPSO, SHADE, sep-CMAES, GA-MPC). The bold red curve corresponds to HPPSO.

Conclusion

This study proposed a novel particle swarm optimization variant named Human Personality-Based Particle Swarm Optimization (HPPSO). The proposed algorithm introduces personality-inspired behavioral diversity into swarm particles to enhance exploration capabilities, particularly in high-dimensional search spaces. Rather than merely proposing another PSO variant, this work aims to introduce a new perspective for modeling heterogeneous behaviors in swarm intelligence algorithms.

Although heterogeneous PSO variants have previously been investigated in the literature, HPPSO introduces heterogeneity through personality-inspired behavioral traits. The objective is not to simulate human psychology accurately, but rather to improve optimization performance by diversifying particle behaviors. Each particle is assigned a personality vector composed of four traits: curiosity, aggressiveness, conformity, and openness.

While curiosity, aggressiveness, and conformity are implemented as behavioral modifications to the standard PSO dynamics, the openness trait introduces a distinctive exploration operator. Specifically, open particles are occasionally attracted toward randomly selected poorly performing particles. Although this mechanism appears counterintuitive, it can assist in escaping local minima by disrupting conventional attraction dynamics and encouraging exploration of previously abandoned regions that may still contain promising solutions.

Empirical results demonstrated that the openness operator improves optimization performance. Preliminary theoretical discussion was provided to explain the effectiveness of this mechanism; however, further analytical investigation is required to better understand its influence on swarm dynamics. The remaining traits were also independently evaluated, and experimental results indicated that they positively contribute to optimization performance.

Although concepts related to curiosity, aggressiveness, and conformity have been discussed separately in previous studies, HPPSO differs by integrating these operators into stochastic personality profiles assigned to individual particles. Consequently, particles exhibit complex and unpredictable

combinations of behaviors, producing swarm dynamics that, to the best of our knowledge, have not been previously explored in the literature.

To further reduce premature convergence and improve adaptability, particles that fail to improve for several iterations undergo personality-vector reinitialization.

Complexity analysis demonstrated that HPPSO does not introduce significant additional computational overhead and maintains approximately the same computational complexity as standard PSO. Furthermore, convergence analysis suggested that HPPSO preserves multiple promising search trajectories rather than collapsing rapidly into a single search region. This diversity preservation appears particularly beneficial in extremely large search spaces, although it may reduce the precision of final solutions. Future work may therefore focus on improving the resolution and refinement of the final search stage.

HPPSO was evaluated using 20 benchmark functions shifted by random vectors to avoid trivial optimization landscapes centered around zero bias. Experiments were conducted in both 30-dimensional and 1000-dimensional settings. The results demonstrated that HPPSO achieved competitive performance in 30-dimensional problems and performed competitively against sep-CMAES, a state-of-the-art optimizer for large-scale optimization, in 1000-dimensional settings.

The algorithm was additionally benchmarked using the CEC2011 test suite, which is particularly suitable for evaluating stochastic optimization methods on highly complex problems. HPPSO outperformed the CEC2011 winner GA-MPC and achieved performance comparable to SHADE and CMAES, which are widely regarded as state-of-the-art optimization algorithms.

Neural network training was selected as a real-world large-scale optimization task to evaluate the effectiveness of HPPSO in massive search spaces. Experimental results on the Digits and Breast Cancer datasets demonstrated that HPPSO is capable of finding competitive solutions within the same number of iterations used by competing algorithms. Moreover, unlike SHADE, HPPSO does not require an external memory archive. In addition, standard CMAES has $O(n^2)$ computational complexity, which becomes increasingly expensive in high-dimensional settings. Although separable CMAES variants alleviate this issue and may outperform HPPSO in certain cases, they do not consistently guarantee the same performance characteristics as full CMAES.

Despite its strong exploration capabilities and stochastic search mechanisms, HPPSO is still not always able to effectively navigate highly multimodal and deceptive optimization landscapes. Addressing this limitation remains an important direction for future research. Additionally, while the current evaluation focused primarily on static optimization benchmarks, further investigation is required to assess the performance of HPPSO in dynamic optimization environments such as robotic path planning.

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